Comprehensive Analysis of Receiver Operating Characteristic (ROC) Curves for Hyperspectral Anomaly Detection

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***Abstract*— The receiver operating characteristic (ROC) curve of detection probability (P_D) versus the false alarm probability (P_F), referred to as 2D ROC curve, has been widely used to evaluate hyperspectral anomaly detection (AD) performance. This article explores several fundamental and conceptual issues of a 2D ROC curve used for AD, which has been overlooked and never investigated in the past. How can a Neyman–Pearson (NP) detector work for AD? How is an ROC curve plotted without probability distributions? Why is a 2D ROC curve reported in the literature as a step function and later linearly interpolated as a linear piecewise function? How can an ROC curve be used to evaluate background suppression (BS)? To address all these issues, a mathematical theory of 2D ROC curve is rederived by a random Neyman–Pearson detector (RNPD) via a threshold parameter τ, which actually determines P_D and P_F, and its detailed theoretical proofs along with comprehensive analysis are also provided. Specifically, a binary communication channel example is included to illustrate how RNPD works. This threshold τ-driven approach, indeed, paves a way for deriving a 3D ROC curve as a function of three parameters: τ, P_D, and P_F. By virtue of a 3D ROC curve, three 2D ROC curves of (P_D,P_F), (P_D,τ), and (P_F,τ) can be derived to perform AD performance analysis effectively in terms of anomaly detectability and BS. Experiments demonstrate that many anomaly detectors that claim to perform well on AD using 2D ROC curves are actually performed very poorly in BS.**

***Index Terms*— Anomaly detection (AD), background suppression (BS), receiver operating characteristic (ROC).**

NOMENCLATURE

AD	Anomaly detection.
ADP	Anomaly detection probability.
ADBS	Anomaly detection with BKG suppression.
AP	Anomaly precision.
AUC	Area under an ROC curve.
BKG	Background.
BD	Background detection.
BDP	Background detection probability.
BS	Background suppression.
CEM	Constrained energy minimization.
CEM-AD	CEM anomaly detector.
DNPD	Deterministic Neyman–Pearson detector.
IOU	Intersect over union.
IS	Insufficient sampling.
JAD	Joint anomaly detection.
JBS	Joint BKG suppression.
NP	Neyman–Pearson.
OA	Overall accuracy.
OADP	Overall anomaly detection probability.
P_D	Detection probability.
P_F	False alarm probability.
PR	Precision rate.
RNPD	Random Neyman–Pearson detector.
RX-AD	Reed–Xiaoli anomaly detector.
ROC	Receiver operating characteristic.
SNPR	Signal-to-noise probability ratio.

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I. INTRODUCTION

A 2D ROC curve is commonly used to evaluate the performance of NP signal detectors [1]. In hyperspectral imaging, the AUC, denoted by $AUC_{(D,F)}$, is generally used for quantitative study and analysis in hyperspectral AD (HAD).

Many target detectors are derived from a likelihood ratio test (LRT) based on the following binary simple hypothesis testing formulation specified by:

$$\begin{aligned} &H_0: \ p_0(\mathbf{r}) \\ &\text{versus} \\ &H_1: \ p_1(\mathbf{r}) \end{aligned} \tag{1}$$

where Γ is the observation space with data samples $\mathbf{r} \in \Gamma$, and $p_0(\mathbf{r})$ and $p_1(\mathbf{r})$ are probability density functions (pdfs) that govern the two hypotheses H_0 and H_1, called the "*null hypothesis*" that represents signal absence and the "*alternative hypothesis*" that represents signal presence, respectively. An optimal detector $\delta(\mathbf{r})$ to solve (1) is shown to be

$$\delta(\mathbf{r}) = \begin{cases} 1; & \text{if } \Lambda(\mathbf{r}) > \tau \\ 0 \text{ or } 1; & \text{if } \Lambda(\mathbf{r}) = \tau \\ 0; & \text{if } \Lambda(\mathbf{r}) < \tau \end{cases} \tag{2}$$

where $\Lambda(\mathbf{r}) = p_1(\mathbf{r})/p_0(\mathbf{r})$ is the LRT of the data sample $\mathbf{r}$ and the threshold τ is determined by prior knowledge, such as prior probabilities of hypotheses and costs. In particular, we define a critical region, $\Gamma_1 = \{\mathbf{r} \in \Gamma | \delta(\mathbf{r}) = 1\}$ consisting of all observations that yield H_1. Then, $\Gamma_0 = \{\mathbf{r} \in \Gamma | \delta(\mathbf{r}) = 0\}$

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consists of all observations that yield H_0, and $\bar{\Gamma} = \{\mathbf{r} \in \Gamma | \delta(\mathbf{r}) = 0 \text{ or } 1\}$ is the boundary region between Γ_0 or Γ_1, which is made up of observations that can go either H_0 or H_1. In order to make (1) applicable to AD, the assumption of the prior knowledge-determined τ is not valid. The NP detection theory provides a feasible solution by imposing a constraint $0 \leq \beta \leq 1$ on $\mathrm{P_F}$, i.e., $\mathrm{P_F}(\delta^{\mathrm{NP}}) \leq \beta$. In this case, there is no need of determining τ. As a result, the detector $\delta(\mathbf{r})$ in (2) becomes an RNPD, $\delta^{\mathrm{RNPD}}(\mathbf{r})$, resulting from the constraint β. Its performance is then evaluated by a curve of $\mathrm{P_D}$ versus $\mathrm{P_F}$, referred to as the 2D ROC curve.

On the other hand, for AD, the target knowledge is unknown. Thus, directly using (1) is not applicable to AD. To be able to take advantage of (1), an alternative approach is to reformulate (1) as a binary composite hypothesis problem from which a generalized LRT (GLRT) can be derived for AD [2], [3], [4], [5], [6]. However, there are several key differences between an RNPD and a GLRT detector. In order for GLRT to take advantage of (1), GLRT makes two assumptions, while an NP does not. One assumption is to assume that there is a secondary dataset available to be used to estimate the unknown target signatures. The other is the pdfs that are Gaussian so that the maximum likelihood estimation is generally used for this purpose. Nevertheless, both RNPDs and GLRT-based detectors arrive at the same evaluation criterion, which is the 2D ROC curve. Unfortunately, several conceptual issues that arise from using such a 2D ROC curve for hyperspectral detection have been overlooked and never investigated in the past.

First, since an RNPD, δ^{RNPD}, must satisfy a constraint imposed by a prescribed $\mathrm{P_F}$ value, β, i.e., $\mathrm{P_F}(\delta^{\mathrm{NP}}) \leq \beta$, δ^{RNPD} is generally a random detector where its randomized detection takes care of the constraint $\mathrm{P_F}(\delta^{\mathrm{NP}}) = \beta$. However, the randomized detection is not needed for GLRT due to the fact that GLRT generally assumes the pdfs to be Gaussian. Thus, when GLRT occurs at the boundary $\bar{\Gamma}$, the probability of the boundary is zero, $P(\bar{\Gamma}) = 0$. This is the reason why the GLRT-derived anomaly detectors reported in the literature never discussed the issue of randomized detections on $\bar{\Gamma}$ [3], [4], [5], [6] because the 2D ROC curve of a GLRT generated from the Gaussian distributions is a continuous and $P(\bar{\Gamma}) = 0$. However, this is not true for hyperspectral imagery where the used hyperspectral data are a discrete set of data samples in which case $P(\bar{\Gamma}) = 0$ is no longer true. That is why we have seen that all ROC curves plotted for AD in the literature are not convex-cap curves but rather linearly piecewise interpolated step functions. Interestingly, to the best of our knowledge, this conceptual issue has been existing, and no one has paid attention to it in the past. As a matter of fact, addressing this issue is very challenging and not as easy as we thought and cannot be taken for granted and lightly.

The second issue is that, once an ROC curve is shown to be a step function calculated from a discrete set of data samples, the follow-up issue is how to calculate the step jumps for a step function-like 2D ROC curve along with its linearly interpolated ROC curve. Such calculations are not trivial or straightforward. It requires the NP lemma [1] to find the probability of randomized detections on $\bar{\Gamma} = \{\mathbf{r} \in \Gamma | \delta(\mathbf{r}) = 0 \text{ or } 1\}$ in (2).

The third issue arises from plotting an ROC curve, which assumes that $\mathrm{P_F}$ is an independent variable and $\mathrm{P_D}$ is a dependent variable. The fact of the matter is that both $\mathrm{P_F}$ and $\mathrm{P_D}$ are actually dependent variables of an implicit independent variable, threshold τ in (2). Thus, both $\mathrm{P_D}$ and $\mathrm{P_F}$ cannot work independently but rather interactively.

The fourth issue is how to plot an ROC curve for a discrete set of data samples. According to the NP detection theory, an ROC curve is plotted for $\mathrm{P_D}$ versus $\mathrm{P_F}$ subject to the constraint, $\mathrm{P_F}(\delta^{\mathrm{NP}}) = \beta$ with $\beta \in [0, 1]$. In other words, an ROC curve is generated by finding the value of $\mathrm{P_D}$ by τ, which is actually determined by $\mathrm{P_F}(\delta^{\mathrm{NP}}) = \beta$. A dilemma arises from the fact that, for a discrete set of data samples $\{\mathbf{r}_i\}_{i=1}^N$, there are only a discrete set of β values, i.e., $\{\beta_i\}_{i=1}^N$ can be generated by $\{\mathbf{r}_i\}_{i=1}^N$ data samples. Thus, for those $\beta \notin \{\beta_i\}_{i=1}^N$, there are no corresponding data samples that can be used to calculate $\mathrm{P_D}$ by imposing the constraint $\mathrm{P_F}(\delta^{\mathrm{NP}}) = \beta$. This implies that using any β to find $\mathrm{P_D}$ does not work. In this case, how the 2D ROC curves reported in the literature are generated and plotted?

To address the above four issues, we reinvent the wheel by considering the detector specified by (2) as an RNPD, which uses the threshold parameter τ instead of β to generate both values of $\mathrm{P_D}$ and $\mathrm{P_F}$ and then plot pairs of ($\mathrm{P_D}$,$\mathrm{P_F}$) to construct an ROC curve. As a consequence, an ROC curve is a step function and can be linearly interpolated by RNPD. In fact, this approach is probably the way that has been used in the literature to generate ROC curves, including MATLAB. However, this threshold τ-to-find $\mathrm{P_F}(\delta^{\mathrm{NP}}) = \beta$ approach has never been validated. To deal with this issue, we interpret an anomaly detector as an RNPD, which makes a random detection on each of data samples and further provide its mathematical proofs and derivations. Most interestingly, this NP-detection approach also gives rise to a concept of how to extend a 2D ROC curve to a 3D ROC curve developed in [7]. The key idea between the traditional 2D ROC curve and a 3D ROC curve lies on how an ROC curve is generated. Comparing to the traditional 2D ROC curve that plots $\mathrm{P_D}$ versus $\mathrm{P_F}$ by varying $\mathrm{P_F}(\delta^{\mathrm{NP}}) = \beta$ as a variable, a 3D ROC curve plots an ROC curve by varying an independent parameter τ as a threshold variable to find pairs of ($\mathrm{P_D}$,$\mathrm{P_F}$) instead of varying $\mathrm{P_F}(\delta^{\mathrm{NP}}) = \beta$ to generate $\mathrm{P_D}$ as the traditional ROC curve is commonly plotted.

Once 2D ROC curve issues are resolved, an inherent issue is its ineffectiveness in measuring the performance of an anomaly detector. Many research efforts have been devoted to designing and developing various AD methods, and claimed that their AD methods were better than others because of their higher values of $\mathrm{AUC_{(D,F)}}$. However, as noted in [2], [7], [8], [9], and [10], AD is quite different from target detection (TD) in several significant aspects. A key difference is that TD is generally carried out with known targets as opposed to AD with unknown targets. As a result, TD is considered to be active signal detection, while AD is considered passive signal detection [11]. A second but very crucial difference is that a target detector is generally designed by a binary simple hypothesis testing formulation with the null hypothesis and alternative hypothesis specified by signal absence and

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signal presence, respectively. Since its major focus is on signal detection, the NP detection theory is used to maximize P_D while constraining P_F at a prescribed value β to reflect the significance level of the detector. The resulting RNPD happens to be LRT proved by the NP lemma. To evaluate its performance, the 2D ROC curve of (P_D,P_F) is particularly derived to measure the effectiveness of an RNPD. In contrast to TD, AD does not know what targets it is looking for. In this case, the binary hypothesis testing formulation in (1) used by TD is not directly applicable to AD. To mitigate this dilemma, the binary simple hypothesis testing formulation is generalized to a binary composite hypothesis testing formulation by making assumption that a target shape/spectrum is known, but its amplitude/abundance is unknown and needs to be estimated. The resultant target detector is called GLRT with detailed derivations provided in [2], [3], [4], [5], [6]. This may also lead to a misconception that $AUC_{(D,F)}$ can be used to evaluate the performance of an anomaly detector in the same way that it is used to evaluate a target detector. Unfortunately, as also shown in recent work in [8] and [9], anomaly detectors can be derived from other aspects, which included GLRT as a special case. This also indicates that blindly using $AUC_{(D,F)}$ to evaluate the performance of an anomaly detector may not be effective. Last but the most critical difference is the BKG effect on AD, which is not considered in TD. However, due to the unavailability of prior knowledge about anomalies, AD is operated in a completely blind environment with no particular targets to be detected. As a result, two scenarios generally occur as anomalies. One is anomalies embedded in BKG in which case anomalies can be unknowingly considered as BKG signal sources. The second scenario is anomalies corrupted by noise in which case anomalies can be mistakenly treated as noise. Unfortunately, most works reported on AD in the literature have been mainly focused on the first scenario but have paid little attention to the second scenario. As shown in [7], [8], [9], and [10], $AUC_{(D,F)}$ generally cannot address the issues of either BS or joint target detectability with BS (ADBS).

By realizing the significant impact of BKG on AD as discussed above, low-rank and sparse representation model- and deep learning-based approaches have been main streams in developing AD algorithms for dealing with BKG in recent years [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27]. In order to show that these methods perform well for AD, $AUC_{(D,F)}$ has been also widely used for this purpose. By judging their produced high values of $AUC_{(D,F)}$, an AD method claims to perform better if its $AUC_{(D,F)}$ value is higher than another. Interestingly, this is, indeed, misleading and unreliable. The main issue arising from $AUC_{(D,F)}$ used for AD hinges on the fact that a 2D ROC curve is derived from the NP detection theory with P_F used as a constraint to calculate P_D. It turns out that many results reported in the literature claimed to perform well but actually did very poorly in BS, such as [19], [20], and [21]. This is because the BKG-based AD methods are designed to take care of the BKG issue, but $AUC_{(D,F)}$ is not effective to evaluate BS. This is similar to a good medicine, which is wrongly prescribed and misused by diagnosis.

The BKG issue is generally not of the main concern in TD. In TD, the target knowledge is known *a priori*, and the traditional 2D ROC curve of (P_D,P_F) is developed to detect targets of interest in communications and signal processing where BKG is simply assumed to be noise. However, in AD, noise and BKG are two crucial factors in AD assessment. Consequently, the 2D ROC curve of (P_D,P_F) is not sufficient to evaluate the performance of AD. In the meantime, the use of $AUC_{(D,F)}$ does not address the BKG issue because both P_D and P_F are determined by the same threshold τ. As a consequence, a higher value of $AUC_{(D,F)}$ yields not only a high value of P_D but also a high value of P_F. Thus, if we interpret P_F as the misprobability of BKG, then a lower value of P_F implies better BS, i.e., a higher value of BS probability, P_{BS}. In this case, we can define $P_{BS} = 1-P_F$. This further indicates that a higher $AUC_{(D,F)}$ also implies a lower value P_{BS} specified by a higher value of P_F, which results in poor BS. In other words, both P_D and P_F cannot function alone and cannot work independently. The key issue arising from the 2D ROC curve is its implicit parameter, threshold τ, which is the one actually determining P_D and P_F. Such a role that τ plays is similar to the latent variable ξ played in deep learning networks.

To resolve the above 2D ROC curve issue, Chang [7] recently developed a 3D ROC curve-based analysis that considered the threshold τ as an independent parameter, while both P_D and P_F can be viewed as dependent parameters through τ. By virtue of a 3D ROC curve, the 2D ROC curve of (P_D,P_F) becomes a special case of a 3D ROC curve. In addition, two new other 2D ROC curves can be also derived as the 2D ROC curve of (P_D,τ) and the 2D ROC curve of (P_F,τ) with their corresponding, $AUC_{(D,\tau)}$ and $AUC_{(F,\tau)}$. Since the 2D ROC curve of (P_D,τ) and the 2D ROC curve of (P_F,τ) are curves plotted by the independent parameter τ, $AUC_{(D,\tau)}$ can be used to evaluate the probability of TD, P_{TD}, and $AUC_{BDP} = 1-AUC_{(F,\tau)}$ can be also used to measure BKG suppressibility as detection probability of BKG, referred to as BDP, individually. This is because $AUC_{(D,F)}$ is calculated by P_D and P_F using the same parameter, threshold τ, it cannot be used to evaluate anomaly detectability or BS alone but rather be used as a compromise between anomaly detectability and BS. It is known that P_{TD} and P_{BDP} are generally not compatible. Thus, we cannot infer BKG suppressibility from anomaly detectability and vice versa. This also explains why $AUC_{(D,F)}$ alone cannot be used to evaluate anomaly detectability and BKG suppressibility for AD independently. In communications and signal processing, BKG suppressibility is simply the noise in signal detection. In this case, $AUC_{(D,F)}$ is developed primarily as a measure to evaluate the effectiveness of a detector but not particularly for anomaly detectability or BKG suppressibility. Therefore, a direct use of $AUC_{(D,F)}$ to measure AD performance as many reported in the literature is not reliable. Unfortunately, such relationship between anomaly detectability and BKG suppressibility has been overlooked and never been explored in AD over the past years. The 3D ROC analysis developed in [7] turns out to be a very powerful and effective evaluation tool to measure AD performance.

Several novelties presented in this article, which cannot be found in [7], are summarized as follows.

1) The most novelty of all is a comprehensive mathematical theory derived from a random NP detection theory for HAD.

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2) The second novelty is to introduce the concept of partitioning a sample data space into three component spaces corresponding to the null hypothesis, alternative hypothesis, and the boundary between these two hypotheses, so as to derive RNPD, in which case the detection performance is determined by the threshold parameter τ instead of probability distributions. As a result, RNPD can be used for discrete datasets with randomized detection determined for unobserved, unavailable, or missing data samples that are unknown and require random decisions. In particular, an ROC curve can be generated by varying a threshold parameter τ rather than varying the $\mathrm{P_F}(\delta^{\mathrm{NP}}) = \beta$ as a variable.
3) The third novelty is to utilize a binary communication channel to rederive an anomaly-BKG hypothesis testing problem so that BKG detection probability can be introduced to address the BKG suppressibility issue. In particular, such a binary communication channel paves a way for how a 2D ROC curve is extended to a 3D ROC curve.
4) The fourth novelty is to reinterpret detection metrics commonly used in computer vision for object/small infrared TD in the context of AUC detection measures.

By virtue of the above novelties, the main contributions can be derived in the following.

1) The first and foremost contribution is to address and provide solutions to several fundamental and conceptual issues arising in plotting an 2D ROC curve, which has been taken for granted but never investigated in the past.
2) The second contribution is to formulate an anomaly detector as an RNPD not GLRT for a discrete set of data samples where RNPD is, indeed, a randomized detector and its solutions along with their theoretical proofs and solutions to show that only detection of given discrete data samples is deterministic, and the detection of unobserved, unavailable, or missing data samples must be random.
3) The third contribution shows that an ROC curve generated by an RNPD for a discrete set of data samples is, indeed, a curve linearly piecewise interpolated by randomized detections on unavailable, unobserved, or missing data samples where the 2D ROC curve is continuous but not differentiable at all the given discrete data samples. On the other hand, if an ROC curve is plotted only for given discrete data samples, the 2D ROC curve is then reduced to a step function and is no longer a linear piece-wise function.
4) The fourth contribution is to use a simple binary communication channel to illustrate the concept of how a randomized RNPD works and the idea of how to plot an accurate ROC curve. This example offers insight into how to formulate an AD problem as an anomaly-BKG detection problem.
5) The fifth contribution is to formulate an AD problem as an anomaly-BKG detection problem so that a confusion matrix-based detection approach using a 3D ROC curve-derived evaluation tool can be developed to replace the traditional ($\mathrm{P_D}$,$\mathrm{P_F}$)-based approach, which can only measure the effectiveness of a detector but not its anomaly detectability and BS. As a result of such a 3D ROC curve, various detection measures can be designed to address AD in anomaly detectability, BKG suppressibility, and joint anomaly detectability-BKG suppressibility.
6) Finally, the last contribution is to derive AUC detection measures counterparts of several detection metrics commonly used in object/small infrared TD and their corresponding AUC detection measures, which may provide more effective criteria for performance analysis.

The remainder of this article is organized as follows. Section II presents an NP detection theory for HAD. Section III develops randomized NP detection theory for discrete sets of data samples. Section IV particularly studies and discusses the issues of plotting a 2D ROC curve. Section V presents a binary communication channel to illustrate how an ROC curve is plotted with great detail. Section VI takes advantage of the binary communication channel discussed in Section V to formulate an AD problem as an anomaly-BKG two detection problem where 3D ROC curve-based detection measures are also derived for HAD. Section VII describes the images to be used for experiments. Section VIII conducts a comprehensive analysis of ROC curves and performs AD evaluation by using 3D ROC-derived detection measures. Section IX draws a conclusion.

II. Neyman–Pearson Detection for Hyperspectral Anomaly Detection

A signal detection can be described as a binary hypothesis testing problem specified by (1). Assume the prior probabilities of hypotheses, $\pi_0 = P(H_0)$, $\pi_1 = P(H_1)$, and the cost functions associated with the hypothesis, c_{ij}, which is the cost of saying H_i when H_j is true. A Bayes detector for (1) can be obtained by

$$\delta^{\mathrm{Bayes}}(\mathbf{r}) = \begin{cases} 1; & \text{if } \Lambda(\mathbf{r}) \geq \tau \\ 0; & \text{if } \Lambda(\mathbf{r}) < \tau \end{cases} \tag{3}$$

where $\Lambda(\mathbf{r}) = p_1(\mathbf{r})/p_0(\mathbf{r})$ is the LRT of the data sample $\mathbf{r}$ as its detector statistic and $\tau = ((\pi_0(c_{10} - c_{00}))/(\pi_1(c_{01} - c_{11})))$ is the threshold. "1" in (3) indicates the data sample $\mathbf{r}$ is detected by $\mathrm{L}(\mathbf{r})$ as a signal sample and "0" otherwise. Using (3), $\mathrm{P_D}$ and $\mathrm{P_F}$ can be found by

$$\mathrm{P_F}(\delta^{\mathrm{Bayes}}) = \int_{\Lambda(\mathbf{r})\geq\tau} p_0(\mathbf{r})d\mathbf{r} \tag{4}$$

$$\mathrm{P_D}(\delta^{\mathrm{Bayes}}) = \int_{\Lambda(\mathbf{r})\geq\tau} p_1(\mathbf{r})d\mathbf{r}. \tag{5}$$

In many real-world applications, the prior knowledge of $\{\pi_0, \pi_1\}$ and c_{ij} is generally not available. Thus, the NP detection theory is developed to address these two main issues. Without making any assumption on $\{\pi_0, \pi_1\}$ and c_{ij}, an NP detection problem is designed to maximize $\mathrm{P_D}$ by constraining $\mathrm{P_F}$ at a prescribed value b as follows:

$$\max_{\delta} \{\mathrm{P_D}(\delta)\} \quad \text{s.t. } \mathrm{P_F}(\delta) \leq \beta, \quad \text{for any given } 0 \leq \beta \leq 1 \tag{6}$$

where β is defined as the significant level of the RNPD. The maximum in (6) is found over all possible detectors, δ.

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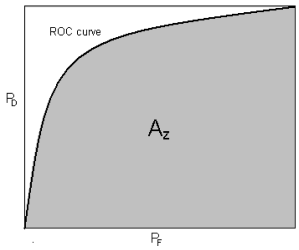


Fig. 1. Example of AUC, A_z, calculated by an ROC curve.

According to the well-known NP lemma [1], a solution to (6) is called an NP detector, δ^{NP}, which can be obtained for each data sample, $\mathbf{r}$, as

$$\delta^{\mathrm{NP}}(\mathbf{r}) = \begin{cases} 1; & \text{if } \Lambda(\mathbf{r}) > \tau^{\beta} \\ 1, & \text{with probability } \gamma(\mathbf{r}) \text{ if } \Lambda(\mathbf{r}) = \tau^{\beta} \\ 0; & \text{if } \Lambda(\mathbf{r}) < \tau^{\beta} \end{cases} \tag{7}$$

where τ used by the Bayes detector in (2) is now replaced by τ^{β}, which is determined by $\mathrm{P_F}$ through the constraint $\mathrm{P_F}(\delta) \leq \beta$ in (6), i.e.,

$$\mathrm{P_F}(\delta^{\mathrm{NP}}) = \int_{\Lambda(\mathbf{r}) \geq \tau^{\beta}} p_0(\mathbf{r}) d\mathbf{r} = \beta \Rightarrow \tau^{\beta} = \mathrm{P_F^{-1}}(\beta). \tag{8}$$

In order to evaluate the performance of an NP detector, a general evaluation tool used in communications and signal processing is the ROC curve, which plots $\mathrm{P_D}$ versus $\mathrm{P_F}$, as shown in Fig. 1, where the AUC, denoted by A_z, is then used as a quantitative measure.

Since the NP detection theory does not require any prior knowledge about hypotheses and costs, the formulation of the NP detection problem in (6) is particularly suitable for AD. However, there are still two main issues in making the NP detection theory applicable to AD. One is the prior knowledge requirement of the signal to be detected. The other is the requirement of pdfs, $p_0(\mathbf{r})$ and $p_1(\mathbf{r})$, to calculate the LRT in (7). Both issues will be addressed in Sections II-A and II-B.

A. Anomaly Detector as GLRT

Over the past years, a traditional approach to AD is to take advantage of the LRT in (2) to derive an anomaly detector. However, there is a challenge for a binary hypothesis testing formulation (1) being used for AD because of how to find LRT without knowing the target to be detected, such as unknown anomalies. To resolve this issue, a classical approach is to extend the binary simple hypothesis testing problem in (1) to a binary composite hypothesis testing formulation so that a GLRT [3], [4], [5], [6] can be also derived [2]. By doing so, two assumptions must be made. One is to assume that there is a secondary dataset available for target signal estimation. The second assumption is that the probability distribution is governed by Gaussian distributions so that LRT can be found for (2). With these two assumptions, (2) was extended to GLRT given by

$$\Lambda(\mathbf{r}) = (\mathbf{r} - \boldsymbol{\mu})^T \mathbf{K}^{-1} (\mathbf{r} - \boldsymbol{\mu}) \tag{9}$$

which is known as RX-AD in [3]. Unfortunately, (9) is still constrained by these two impractical assumptions because the hyperspectral noise is generally not Gaussian, and the availability of a second dataset is not realistic. In a recent work [8], Chang showed that a general anomaly detector (GAD) could be derived as

$$\delta^{\mathrm{GAD}}(\mathbf{r}) = \kappa(\mathbf{r} - \boldsymbol{\mu})^T \mathbf{K}^{-1} (\mathbf{r} - \boldsymbol{\mu}) \quad \text{for some scalar constant } \kappa \tag{10}$$

from signal-to-noise ratio (SNR) or spectral angle without appealing for the binary hypothesis testing problem in (1). By setting $\kappa = 1$ in (10), GAD is reduced to RX-AD in [3]. Using the form of (2), GAD can be specified by

$$\delta^{\mathrm{GAD}}(\mathbf{r}) = \begin{cases} 1; & \text{if } \Lambda(\mathbf{r}) = \kappa(\mathbf{r} - \boldsymbol{\mu})^T \mathbf{K}^{-1} (\mathbf{r} - \boldsymbol{\mu}) \geq \tau \\ 0; & \text{if } \Lambda(\mathbf{r}) = \kappa(\mathbf{r} - \boldsymbol{\mu})^T \mathbf{K}^{-1} (\mathbf{r} - \boldsymbol{\mu}) < \tau. \end{cases} \tag{11}$$

Once again, Chang also used a similar SNR approach to show in [9] that the well-known CEM in [28] and [29] can be further extended to a CEM-AD given by

$$\Lambda(\mathbf{r}) = \kappa \mathbf{r}^T \mathbf{R}^{-1} \mathbf{r}. \tag{12}$$

Now, replacing $\Lambda(\mathbf{r}_i)$ in (11) with (12), CEM-AD can be specified by

$$\delta^{\text{CEM-AD}}(\mathbf{r}) = \begin{cases} 1; & \text{if } \Lambda(\mathbf{r}) = \kappa \mathbf{r}^T \mathbf{R}^{-1} \mathbf{r} \geq \tau \\ 0; & \text{if } \Lambda(\mathbf{r}) = \kappa \mathbf{r}^T \mathbf{R}^{-1} \mathbf{r} < \tau. \end{cases} \tag{13}$$

By letting $\kappa = 1$ in (13), CEM-AD is reduced to R-AD in [11], [30], [31], and [32].

B. RNPD With No Need for Probability Distributions

Since the pdfs, $p_0(\mathbf{r}_i)$ and $p_1(\mathbf{r}_i)$, are generally not available for HAD, directly using LRT, $\Lambda(\mathbf{r}_i) = p_1(\mathbf{r}_i)/p_0(\mathbf{r}_i)$ in (2), is not applicable. To mitigate this dilemma, we focus on the detector statistics specified by LRT and reinterpret the LRT in (2) as an anomaly detector, which can be obtained by the threshold τ without pdfs to calculate $\Lambda(\mathbf{r}_i) = p_1(\mathbf{r}_i)/p_0(\mathbf{r}_i)$ in (2). To do so, for each specified $\mathrm{P_F}(\delta^{\mathrm{NP}}) = \beta$, we define a critical region, Γ_1^{β}, via (7) by

$$\Gamma_1^{\beta} = \left\{ \mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) > \tau^{\beta} \right\}. \tag{14}$$

Then,

$$\Gamma_0^{\beta} = \left\{ \mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) < \tau^{\beta} \right\} \tag{15}$$

$$\bar{\Gamma}^{\beta} = \left\{ \mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) = \tau^{\beta} \right\} \tag{16}$$

where $\bar{\Gamma}$ is the boundary between Γ_1^{β} and Γ_0^{β} with $\Gamma = \Gamma_1^{\beta} \cup \Gamma_0^{\beta} \cup \bar{\Gamma}^{\beta}$.

By virtue of (14)–(16), the RNPD in (7) works without prior knowledge of $p_0(\mathbf{r})$ and $p_1(\mathbf{r})$ but rather its detector statistics, $\Lambda(\mathbf{r})$, which is not necessary to be LRT. Consequently, the RNPD in (7) can be reinterpreted as

$$\delta^{\mathrm{NP}}(\mathbf{r}) = \begin{cases} 1; & \text{if } \mathbf{r} \in \Gamma_1^{\beta} \\ 1, & \text{with probability } \gamma(\mathbf{r}) \text{ if } \mathbf{r} \in \bar{\Gamma}^{\beta} \\ 0; & \text{if } \mathbf{r} \in \Gamma_0^{\beta} \end{cases} \tag{17}$$

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for each data sample $\mathbf{r}$ with the constraint specified by (8). With assumption that $\Gamma_1^\beta \cap \bar{\Gamma}^\beta = \emptyset$, $\mathrm{P_F}$ in (5) and $\mathrm{P_D}$ in (5) can be reexpressed via (17) as

$$\mathrm{P_F}(\delta^{\mathrm{NP}}(\mathbf{r})) = P_0\left(\Gamma_1^\beta\right) + \gamma(\mathbf{r}) P_0(\bar{\Gamma}^\beta) \tag{18}$$

$$\mathrm{P_D}(\delta^{\mathrm{NP}}(\mathbf{r})) = P_1\left(\Gamma_1^\beta\right) + \gamma(\mathbf{r}) P_1(\bar{\Gamma}^\beta) \tag{19}$$

which, in fact, are the base of why linear interpolation can be used to interpret unavailable data samples not provided by a given discrete set of data samples.

As a matter of fact, the randomized detector in (17) can be really obtained by two DNPDs, $\delta^{\mathrm{DNPD}}_{\Gamma_1^\beta}(\mathbf{r})$ and $\delta^{\mathrm{DNPD}}_{\Gamma_1^\beta \cup \bar{\Gamma}^\beta}(\mathbf{r})$, given by

$$\delta^{\mathrm{DNPD}}_{\Gamma_1^\beta}(\mathbf{r}) = \begin{cases} 1; & \text{if } \mathbf{r} \in \Gamma_1^\beta \\ 0; & \text{if } \mathbf{r} \in \Gamma_0^\beta \cup \bar{\Gamma}^\beta \end{cases} \tag{20}$$

and

$$\delta^{\mathrm{DNPD}}_{\Gamma_1^\beta \cup \bar{\Gamma}^\beta}(\mathbf{r}) = \begin{cases} 1; & \text{if } \mathbf{r} \in \Gamma_1^\beta \cup \bar{\Gamma}^\beta \\ 0; & \text{if } \mathbf{r} \in \Gamma_0^\beta \end{cases} \tag{21}$$

so that $\delta^{\mathrm{NP}}(\mathbf{r})$ in (17) can be considered as a linear mixture of $\delta^{\mathrm{DNPD}}_{\Gamma_1^\beta}(\mathbf{r})$ and $\delta^{\mathrm{DNPD}}_{\Gamma_1^\beta \cup \bar{\Gamma}^\beta}(\mathbf{r})$ as RNPD denoted by $\delta^{\mathrm{RNPD}}_{\Gamma_1^\beta \cup \gamma \bar{\Gamma}^\beta}(\mathbf{r})$ with the random parameter $\gamma(\mathbf{r})$ given by

$$\delta^{\mathrm{RNPD}}_{\Gamma_1^\beta \cup \gamma \bar{\Gamma}^\beta}(\mathbf{r}) = (1 - \gamma(\mathbf{r}))\delta^{\mathrm{DNPD}}_{\Gamma_1^\beta}(\mathbf{r}) + \gamma(\mathbf{r})\delta^{\mathrm{DNPD}}_{\Gamma_1^\beta \cup \bar{\Gamma}^\beta}(\mathbf{r}) \tag{22}$$

where $\gamma(\mathbf{r})$ is exactly the probability in (17) to declare H_1 when $\mathbf{r} \in \bar{\Gamma}^\beta$.

Interestingly, all the anomaly detectors reported in the literature were either (20) or (21) but never specified and discussed what types of anomaly detectors were used. It is obvious that using (20) and (21) yields different results as detailed in [7]. As a matter of fact, due to the use of a discrete set of data samples, a hyperspectral anomaly detector is, indeed, a randomized detector $\delta^{\mathrm{RNPD}}_{\Gamma_1^\beta \cup \gamma \bar{\Gamma}^\beta}(\mathbf{r})$ described by (22), which is a linear random mixture of two DNPDs [see (20) and (21)]. This article is believed to be the first work ever reported in using (22) as a randomized anomaly detector for hyperspectral imagery.

III. Randomized RNPD Resulting From a Discrete Set of Data Samples

According to (7) and (17), an RNPD is generally a randomized detector. For a continuous dataset Γ, the probability of the boundary $\bar{\Gamma}$ between Γ_0 and Γ_1, $P_1(\bar{\Gamma}) = P_0(\bar{\Gamma}) = 0$, is zero, which implies $\gamma(\mathbf{r}) = 0$. Thus, both $\delta_1^{\mathrm{NP}}(\mathbf{r})$ in (20) and $\delta_2^{\mathrm{NP}}(\mathbf{r})$ in (21) yield the same detector for $\mathbf{r}$. Unfortunately, this is no longer true for a discrete dataset Γ because only a discrete set of false alarm probabilities can be calculated for an ROC curve. For each data sample $\mathbf{r}$, the two detectors, $\delta^{\mathrm{NP}}_{\Gamma_1^\beta}(\mathbf{r})$ in (20) and $\delta^{\mathrm{NP}}_{\Gamma_1^\beta \cup \bar{\Gamma}^\beta}(\mathbf{r})$ in (21), can be designed, each of which gives different values of $\mathrm{P_D}$ and $\mathrm{P_F}$. To take care of both detectors, the optimal detector should be a randomized detector $\delta^{\mathrm{RNPD}}_{\Gamma_1^\beta \cup \gamma \bar{\Gamma}^\beta}(\mathbf{r})$ specified by (22). Furthermore, since the values of $\mathrm{P_D}$ and $\mathrm{P_F}$ are not available between two data samples, it is very important to appropriately interpolate the missing values between two data samples. In what follows, we will show how an RNPD, $\delta^{\mathrm{RNPD}}_{\Gamma_1^\beta \cup \gamma \bar{\Gamma}^\beta}(\mathbf{r})$, to interpret unavailable data samples from a discrete set of data samples via its randomized detections on data samples without using probability distributions.

A. Step Functions of ROC Curves Resulting From Discrete Sets of Data Samples

As noted, a key issue in plotting ROC curves arises from the fact that the dataset is discrete. In other words, when the dataset Γ is continuous, for any given $P_{\mathrm{F}}(\delta^{\mathrm{NP}}) = \beta$, we can always find τ^β through (8) to find $P_{\mathrm{F}}^{-1}(\beta) = \tau^\beta$ that can be used to calculate $\mathrm{P_D}(\delta^{\mathrm{NP}}) = \int_{\Lambda(\mathbf{r}) \geq \tau^\beta} p_1(\mathbf{r}) d\mathbf{r}$. However, when the set of data samples, $\Gamma = \{\mathbf{r}_i\}_{i=1}^N$, is discrete, there only $\{\Lambda(\mathbf{r}_i)\}_{i=1}^N$ can produce strictly decreasing $\mathrm{P_F}$ values, $\{\beta_i\}_{i=1}^N$ with β_i calculated by $\Lambda(\mathbf{r}_i)$, and $\beta_{i+1} < \beta_i$. Thus, for $\beta \notin \{\beta_i\}_{i=1}^N$, there is no corresponding data sample $\mathbf{r}$ in $\Gamma = \{\mathbf{r}_i\}_{i=1}^N$ that can produce β to plot its corresponding point ($\mathrm{P_D}$,$\mathrm{P_F}$) on the 2D ROC curve. This implies that the generated ROC curve should be a step function. However, according to most of the ROC curves reported in the literature, they are actually linear piecewise continuous functions. How could that be? It is a very fundamental issue but has never been investigated in the hyperspectral detection literature. This section shows that such linearly interpolated piecewise ROC curve is, indeed, generated by an RNPD as a randomized detector described in the following theorem where the proof follows an idea similar to the proof of the NP lemma given in [1, pp. 23–25].

Theorem 1: For a given discrete set of data samples, $\Gamma = \{\mathbf{r}_i\}_{i=1}^N$, a 2D ROC curve is, in fact, a step function where each data sample $\mathbf{r}_i$ in $\Gamma = \{\mathbf{r}_i\}_{i=1}^N$ can produce $\mathrm{P_F}(\Lambda(\mathbf{r}_i)) = \beta_i$ via the detector statistics, $\Lambda(\mathbf{r}_i)$. However, for $\beta \notin \{\beta_i\}_{i=1}^N$, we can use its $\delta^{\mathrm{NP}}(\mathbf{r})$ in (17) as a randomized detector, $\delta^{\mathrm{RNPD}}_{\Gamma_1^\beta \cup \gamma \bar{\Gamma}^\beta}(\mathbf{r})$ to interpolate an unavailable data sample $\mathbf{r}$ that produces $\mathrm{P_F}(\Lambda(\mathbf{r})) = \beta$ with

$$\gamma(\mathbf{r}) P_1(\bar{\Gamma}^\beta) \tag{23}$$

where

$$\gamma(\mathbf{r}) = \frac{\beta - P_0\left(\Gamma_1^\beta\right)}{P_0(\bar{\Gamma}^\beta)} \tag{24}$$

and τ^β is the smallest value τ in (14) such that

$$P_0\left(\Gamma_1^\beta\right) = P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) > \tau\}) \leq \beta \tag{25}$$

i.e., $\tau^\beta = \arg\min_\tau \{P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) > \tau\}) \leq \beta\}$ with

$$\Gamma_1^\beta = \left\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) > \tau^\beta\right\} \tag{26}$$

$$\bar{\Gamma}^\beta = \left\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) = \tau^\beta\right\}. \tag{27}$$

Proof: According to (18), the constraint of $\mathrm{P_F}(\delta^{\mathrm{NP}}) = \beta_i$ at the ith data sample $\mathbf{r}_i$ is given by

$$\mathrm{P_F}(\delta^{\mathrm{NP}}) = P_0(\Gamma_1) + \gamma(\mathbf{r}_i) P_0(\bar{\Gamma}) = \beta_i \Rightarrow \gamma(\mathbf{r}_i) = 0 \tag{28}$$

because of $P_0(\Gamma_1) = \beta_i$.

However, if $\beta \notin \{\beta_i\}_{i=1}^N$, then

$$\mathrm{P_F}(\delta^{\mathrm{NP}}) = P_0(\Gamma_1) + \gamma(\mathbf{r}) P_0(\bar{\Gamma}) = \beta \Rightarrow \gamma(\mathbf{r}) = \frac{\beta - P_0(\Gamma_1)}{P_0(\bar{\Gamma})} \quad \text{provdied that } \bar{\Gamma} \neq \emptyset \tag{29}$$

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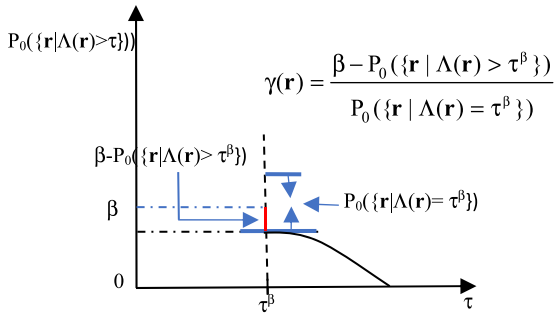


Fig. 2. Threshold and randomization for a β-level RNPD.

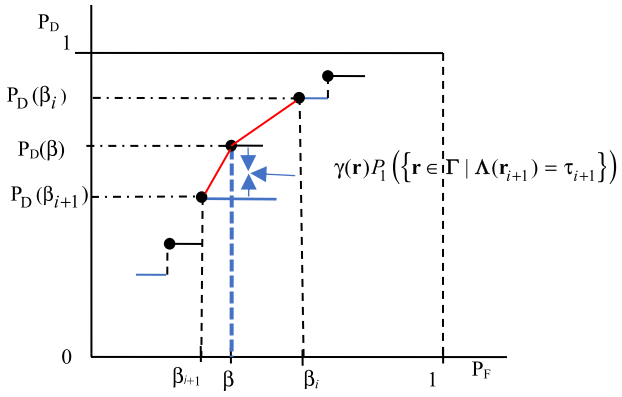


Fig. 3. Step jump at β_{i+1} with $\beta_{i+1} < \beta < \beta_i$.

where $\gamma(\mathbf{r})$ in (29) is the probability of declaring $\mathbf{r}$ as a signal sample when $\mathbf{r} \in \bar{\Gamma}$. More specifically, using the definition of τ^{β} in (25)–(27), (28) can be expressed as

$$\gamma(\mathbf{r}) = \frac{\beta - P_0\left(\Gamma_1^{\beta}\right)}{P_0(\bar{\Gamma}^{\beta})} = \frac{\beta - P_0\left(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) > \tau^{\beta}\}\right)}{P_0\left(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) = \tau^{\beta}\}\right)} \tag{30}$$

as shown in Fig. 2.

From (30), $\mathrm{P_D}$ of $\mathbf{r}$ can be calculated by

$$\mathrm{P_D}(\beta) \equiv \mathrm{P_D}\left(\delta^{\mathrm{RNPD}}_{\Gamma_1^{\beta} \cup_{\gamma} \bar{\Gamma}^{\beta}}(\mathbf{r})\right) = P_1\left(\Gamma_1^{\beta}\right) + \gamma(\mathbf{r}) P_1(\bar{\Gamma}^{\beta}) \tag{31}$$

where the second term of (31) is precisely the step jump given by (23). ■

Theorem 1 shows that an RNPD at any unavailable data sample $\mathbf{r}$ between $\Gamma = \{\mathbf{r}_i\}_{i=1}^{N}$ is, indeed, a randomized detector $\delta^{\mathrm{RNPD}}_{\Gamma_1^{\beta} \cup_{\gamma} \bar{\Gamma}^{\beta}}(\mathbf{r})$ with $\gamma(\mathbf{r}) > 0$ given by (24). It also explains why all ROC curves reported in the literature are linearly interpolated curves that result from a randomized detector $\gamma(\mathbf{r})$ in (30). Theorem 1 shows that, for each individual hyperspectral image pixel, its anomaly detector is a deterministic detector, but a randomized detector on any unavailable data sample is actually linearly interpolated by (29).

It is worth noting that $\{\tau_i\}_{i=1}^{N}$ are strictly increasing threshold values as opposed to their corresponding $\mathrm{P_F}$ values, $\{\beta_i\}_{i=1}^{N}$, which are, indeed, strictly decreasing values. This implies that, for any unavailable data sample $\mathbf{r}$ between $\mathbf{r}_i$ and $\mathbf{r}_{i+1}$, $\delta^{\mathrm{DNPD}}(\mathbf{r})$ is a randomized detector with its step jump given by $\gamma(\mathbf{r}) P_1(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) = \tau_{i+1}\})$, as shown in Fig. 3. In other words, a hyperspectral anomaly detector is, in fact, a randomized detector, which makes random detections by (24).

Using Fig. 3, we can calculate the AUC of a 2D ROC curve of $(\mathrm{P_D}, \mathrm{P_F})$, $\mathrm{AUC}_{(\mathrm{D,F})}$, by the following theorem.

Theorem 2:

$$\begin{aligned} \mathrm{AUC}^{\mathrm{true}}_{\Gamma_1}(\mathrm{D,F}) &= \mathrm{AUC}^{\mathrm{DNPD}}_{\Gamma_1^{\beta}}(\mathrm{D,F}) \\ &= \sum_{i=1}^{N-1} P_1(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) = \tau_i\})(\beta_i - \beta_{i+1}) \\ &= \sum_{i=1}^{N-1} \mathrm{P_D}(\beta_i)(\beta_i - \beta_{i+1}) \end{aligned} \tag{32}$$

for $\delta^{\mathrm{DNPD}}_{\Gamma_1^{\beta}}(\mathbf{r})$ in (20) with "DNPD" used to emphasize that it is a deterministic NPD

$$\begin{aligned} \mathrm{AUC}^{\mathrm{true}}_{\Gamma_1 \cup \bar{\Gamma}}(\mathrm{D,F}) &= \mathrm{AUC}^{\mathrm{DNPD}}_{\Gamma_1^{\beta} \cup \bar{\Gamma}^{\beta}}(\mathrm{D,F}) \\ &\quad \times \sum_{i=1}^{N-1} P_1(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) = \tau_{i+1}\})(\beta_i - \beta_{i+1}) \\ &= \sum_{i=1}^{N-1} \mathrm{P_D}(\beta_{i+1})(\beta_i - \beta_{i+1}) \end{aligned} \tag{33}$$

for $\delta^{\mathrm{DNPD}}_{\Gamma_1^{\beta} \cup \bar{\Gamma}^{\beta}}(\mathbf{r})$ in (21) with "DNPD" used to emphasize that it is a deterministic NPD.

Furthermore, for any $\mathrm{P_F}(\mathbf{r}_i) = \beta_{i+1} < \mathrm{P_F}(\mathbf{r}) = \beta < \mathrm{P_F}(\mathbf{r}_{i+1}) = \beta_i$ of unavailable data samples $\mathbf{r}$ between $\mathbf{r}_i$ and $\mathbf{r}_{i+1}$, i.e., $\mathbf{r} \notin \{\mathbf{r}_i\}_{i=1}^{N}$, its $\mathrm{P_D}(\beta)$ is linearly interpolated by an RNPD, $\delta^{\mathrm{RNPD}}_{\Gamma_1^{\beta} \cup_{\gamma} \bar{\Gamma}^{\beta}}(\mathbf{r})$ via (22) as

$$\mathrm{P_D}(\beta) = (1 - \gamma(\mathbf{r}))\mathrm{P_D}(\beta_i) + \gamma(\mathbf{r})\mathrm{P_D}(\beta_{i+1}) \tag{34}$$

with $\gamma(\mathbf{r})$ given by

$$\gamma(\mathbf{r}) = \frac{\beta - P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) > \tau^{\beta}\})}{P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) = \tau^{\beta}\})} > 0 \tag{35}$$

and its AUC is given by

$$\begin{aligned} \mathrm{AUC}^{\mathrm{RNPD}}_{\Gamma_1 \cup_{\gamma} \bar{\Gamma}}(\mathrm{D,F}) &= \mathrm{AUC}^{\mathrm{RNPD}}_{\Gamma_1 \cup_{\gamma} \bar{\Gamma}}(\mathrm{D,F}) \\ &= \sum_{i=1}^{N-1} ((1 - \gamma(\mathbf{r}))\mathrm{P_D}(\beta_i) + \gamma(\mathbf{r})\mathrm{P_D}(\beta_{i+1})) \\ &\quad \times (\beta_i - \beta_{i+1}) \end{aligned} \tag{36}$$

which is right between $\mathrm{AUC}^{\mathrm{true}}_{\Gamma_1}(\mathrm{D,F}) = \mathrm{AUC}\delta^{\mathrm{DNPD}}_{\Gamma_1}$ in (32) and $\mathrm{AUC}^{\mathrm{DNPD}}_{\Gamma_1 \cup \bar{\Gamma}}(\mathrm{D,F}) = \mathrm{AUC}(\delta^{\mathrm{RNPD}}_{\Gamma_1 \cup \bar{\Gamma}})$ in (33), i.e.,

$$\mathrm{AUC}^{\mathrm{DNPD}}_{\Gamma_1}(\mathrm{D,F}) < \mathrm{AUC}^{\mathrm{RNPD}}_{\Gamma_1 \cup_{\gamma} \bar{\Gamma}}(\mathrm{D,F}) < \mathrm{AUC}^{\mathrm{DNPD}}_{\Gamma_1 \cup \bar{\Gamma}}(\mathrm{D,F}).$$

Proof: Since the dataset is $\Gamma = \{\mathbf{r}_i\}_{i=1}^{N}$, and we assume that $\Lambda(\mathbf{r}_i) = \tau_i < \Lambda(\mathbf{r}_i) = \tau_{i+1}$ without loss of generality, $\mathrm{P_F}(\mathbf{r}_i) = \beta_i > \mathrm{P_F}(\mathbf{r}_{i+1}) = \beta_{i+1}$. According to (25), ">" in $P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) > \tau^{\beta}\}) \leq \beta$ must be "*strictly*" greater than. Since $\beta_{i+1} < \beta < \beta_i$, this implies that τ^{β} must be $\tau^{\beta} = \Lambda(\mathbf{r}_{i+1}) = \tau_{i+1}$ which is the smallest value to satisfy (25). Thus, from (30) and Fig. 2, $\gamma(\mathbf{r})$ can be obtained by (35). Also, the details of the randomized detections at the unavailable data samples $\mathbf{r}$ between $\mathbf{r}_i$ and $\mathbf{r}_{i+1}$ are illustrated in Fig. 3 where the data sample $\mathbf{r}$ can produce $\mathrm{P_F} = \beta$ and $\mathrm{P_D}(\beta) = \mathrm{P_D}(\beta_i) + \gamma(\mathbf{r}) P_1(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) = \tau_{i+1}\})$ for any β in the range of $\beta_{i+1} < \beta < \beta_i$.

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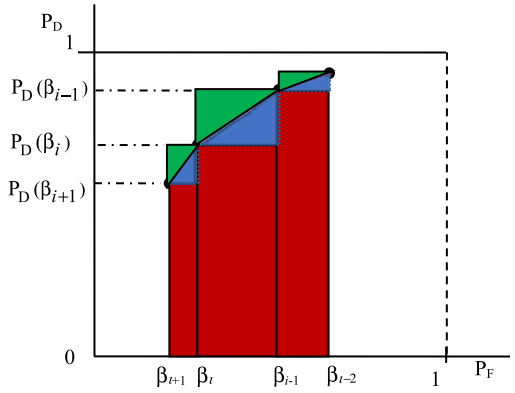


Fig. 4. AUC calculated by $\mathrm{AUC}(\delta^{\mathrm{DNPD}}_{\Gamma_1})$, $\mathrm{AUC}(\delta^{\mathrm{RNPD}}_{\Gamma_1\cup\gamma\bar{\Gamma}})$, and $\mathrm{AUC}(\delta^{\mathrm{DNPD}}_{\Gamma_1\cup\bar{\Gamma}})$ on $\beta_{i+1} < \beta_i < \beta_{i-2}$.

Using (34) and (35), we can calculate $\mathrm{AUC}_{(\mathrm{D,F})}$ by (36). ■

By virtue of Theorem 2, we can calculate $\mathrm{AUC}_{(\mathrm{D,F})}$ based on three detectors: $\delta^{\mathrm{DNPD}}_{\Gamma_1^\beta}(\mathbf{r})$ in (20), $\delta^{\mathrm{DNPD}}_{\Gamma_1^\beta}(\mathbf{r})$ in (21), and $\delta^{\mathrm{RNPD}}_{\Gamma_1^\beta\cup\gamma\bar{\Gamma}^\beta}(\mathbf{r})$ in (22). That is,

$$\mathrm{AUC}^{\mathrm{true}}_{\Gamma_1}(\mathrm{D,F}) = \mathrm{AUC}^{\mathrm{DNPD}}_{\Gamma_1}(\mathrm{D,F}) = \sum_{i=1}^{N-1} \mathrm{P_D}(\beta_i)(\beta_i - \beta_{i+1}) \tag{37}$$

$$\mathrm{AUC}^{\mathrm{true}}_{\Gamma_1\cup\bar{\Gamma}}(\mathrm{D,F}) = \mathrm{AUC}^{\mathrm{DNPD}}_{\Gamma_1\cup\bar{\Gamma}}(\mathrm{D,F}) = \sum_{i=1}^{N-1} \mathrm{P_D}(\beta_{i+1}) \times(\beta_i - \beta_{i+1}) \tag{38}$$

$$\mathrm{AUC}^{\mathrm{RNPD}}_{\Gamma_1\cup\gamma\bar{\Gamma}}(\mathrm{D,F}) = \mathrm{AUC}^{\mathrm{RNPD}}_{\Gamma_1\cup\gamma\bar{\Gamma}}(\mathrm{D,F}) = \sum_{i=1}^{N-1} ((1-\gamma(\mathbf{r}))\mathrm{P_D}(\beta_i) + \gamma(\mathbf{r})\mathrm{P_D}(\beta_{i+1})) \times(\beta_i - \beta_{i+1}). \tag{39}$$

Fig. 4 shows the AUC values calculated by $\delta^{\mathrm{NP}}_{\Gamma_1^\beta}(\mathbf{r})$, $\delta^{\mathrm{NP}}_{\Gamma_1^\beta}(\mathbf{r})$, and $\delta^{\mathrm{RNPD}}_{\Gamma_1^\beta\cup\gamma\bar{\Gamma}^\beta}(\mathbf{r})$ using (37)–(37) only on the range of $\beta_{i+1} < \beta_t < \beta_{i-1}$. It should be noted that, in Fig. 4, the red-shaded areas calculated by $\mathrm{AUC}^{\mathrm{DNPD}}_{\Gamma_1}(\mathrm{D,F}) = \sum_{j=-1}^{1} \mathrm{P_D}(\beta_{i-j})(\beta_{i-j} - \beta_{i-j-1})$ using (37), the area calculated by $\mathrm{AUC}^{\mathrm{DNPD}}_{\Gamma_1\cup\bar{\Gamma}}(\mathrm{D,F}) = \sum_{j=-1}^{1} \mathrm{P_D}(\beta_{i-j})(\beta_{i-j} - \beta_{i-j-1})$ using (38), are the red-shaded rectangular areas plus the triangle areas shaded by blue and green, and the AUC value calculated by $\mathrm{AUC}^{\mathrm{DNPD}}_{\Gamma_1\cup\gamma\bar{\Gamma}}(\mathrm{D,F}) = (1-\gamma)\mathrm{AUC}(\delta^{\mathrm{DNPD}}_{\Gamma_1}) + \gamma\,\mathrm{AUC}(\delta^{\mathrm{DNPD}}_{\Gamma_1\cup\bar{\Gamma}}))$ using (39) is right between these two AUC values, which is the red-shaded rectangular areas plus the blue-shaded triangle areas.

Three remarks are noteworthy.

1) Finding $P_0(\{\mathbf{r}\in\Gamma|\Lambda(\mathbf{r}) > \tau^\beta\}) \le \beta$ in (35) requires a numerical algorithm to solve τ^β, which generally results in numerical errors. Consequently, one great benefit of Theorem 2 is to show that τ^β can be found by $\tau^\beta = \Lambda(\mathbf{r}_{i+1}) = \tau_{i+1}$ in (33) for any β in $\beta_{i+1} < \beta < \beta_i$ without appealing for any numerical algorithm to calculate $\gamma(\mathbf{r})$ in (34).

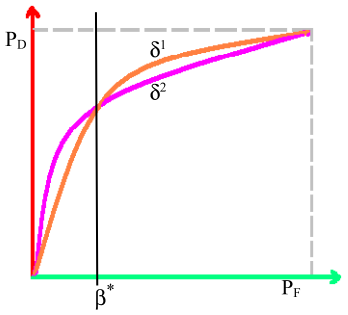


Fig. 5. Example of ROC curves generated by two detectors with the same A_z where β^* is $\mathrm{P_F}$ corresponding to the intersection point of the two ROC curves.

2) The commonly ROC curves reported in the literature, including MATLAB, are, indeed, generated by linear interpolation given by (35) and (36).
3) Thus far, to the best of our knowledge, the works reported on ROC analysis in HAD had overlooked at this fundamental and conceptual issue by taking their generated ROC curves as step function-linearly interpolated curves for granted. The fact of the matter is that this issue is not trivial and cannot be discarded. For a discrete set of data samples, an RNPD must be used instead of GLRT because an RNPD is a randomized detection, which can be used to linearly interpolate the unavailable data samples among $\Gamma = \{\mathbf{r}_i\}_{i=1}^N$. This results in ROC curves as step functions, which can be linearly interpolated by an NP randomized detector. On the other hand, for a continuous set of data samples, an RNPD is a deterministic detector since $\gamma(\mathbf{r}_i)$ in (17) is zero due to $P_0(\bar{\Gamma}) = 0$. This gives rise to a continuous convex-cap ROC curve via (8). For a GLRT that makes an assumption on Gaussian distribution where the pdfs $p_1(\mathbf{r})$ and $p_0(\mathbf{r})$ are known and Gaussian, the 2D ROC curve produced by its RNPD is a continuous convex-cap function, and the detector turns out to be the Mahalanobis distance in (9). However, in a GLRT, such as (11) and (12), without the pdfs $p_1(\mathbf{r})$ and $p_0(\mathbf{r})$, it must deal with the discrete set of data samples, $\Gamma = \{\mathbf{r}_i\}_{i=1}^N$. This can be only done by an RNPD. As a consequence, its generated 2D ROC curve can be found and calculated by Theorem 2.

B. Ineffectiveness in Evaluation for BKG Suppression

Since RX-AD was derived from the binary hypothesis testing formulation, it is natural to use the 2D ROC curve of $(\mathrm{P_D},\mathrm{P_F})$ to evaluate its detection performance. However, a common issue arising from $\mathrm{AUC}_{(\mathrm{D,F})}$ is that it can be only used to evaluate a target detector. It cannot be used to evaluate target detectability and BKG suppressibility. For example, on some occasions, two detectors δ^1 and δ^2 may generate two different 2D ROC curves but have the identical value of $\mathrm{AUC}_{(\mathrm{D,F})}$, as shown in Fig. 5, where the vertical line intersects their ROC curves with the cross point corresponding to $\mathrm{P_F} = \beta^*$. In this case, δ^2 performs better than δ^1 in BS

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when $P_F < \beta^*$. On the other hand, δ^1 performs better than δ^2 in P_D when $P_F \geq \beta^*$. This is particularly true in many results reported in the literature where the compared AD methods produced nearly the same $AUC_{(D,F)}$ values within a range in difference less than 10^{-2}. In this case, how can we claim one is better than another, not mentioning the fact that these detectors could also produce $AUC_{(D,F)}$ values in other datasets with opposite results. Thus, the issue is "how can we evaluate these detectors and in what sense?" The 2D ROC curve fails to resolve this issue because the $AUC_{(D,F)}$ of both detectors yields the same identical result.

Apparently, some recently published works have realized this issue and have taken an analysis approach by evaluating P_D by fixing P_F at a certain value, for example, 10^{-2} or comparing P_F by fixing P_D at 100% [20, Tables II and III]. However, this is not reliable because each pair of (P_D,P_F) only shows up a single point on the 2D ROC curve and cannot address the entire issue.

C. Imprecise ROC Plots

The issue of using (4) to plot ROC curves lies on the use of $\Lambda(\mathbf{r}_i) = p_1(\mathbf{r}_i)/p_0(\mathbf{r}_i)$ as LRT to calculate P_F in (4) and P_D in (5), both of which require the knowledge of $p_0(\mathbf{r})$ and $p_1(\mathbf{r})$. This leads to one key issue, how to deal with a discrete set of data samples, $\Gamma = \{\mathbf{r}_i\}_{i=1}^N$ to plot ROC curve. According to theoretical derivations for the NP detection theory derived from (1) to (8), for a continuous set of data samples, Γ, an ROC curve must be convex-cap with starting at the point (P_D,P_F) = (0,0) and ending at the point (P_D,P_F) = (1,1). However, for a discrete set of data samples, Γ, this is not true. It is particularly the case for hyperspectral imagery where the number of hyperspectral pixel vectors is finite, that is, $\Gamma = \{\mathbf{r}_i\}_{i=1}^N$ is finite and discrete. For example, as shown in [7], if the detector $\delta^{\mathrm{NP}}_{\Gamma_1^\beta}(\mathbf{r})$ specified by (20) is used, the 2D ROC curve will start at (0,0) but will not end at (1,1). On the other hand, if the detector $\delta^{\mathrm{NP}}_{\Gamma_1^\beta \cup \bar{\Gamma}^\beta}(\mathbf{r})$ specified by (21) is used, the 2D ROC curve will not start at (0,0) but will end at (1,1). In either case, a 2D ROC curve never satisfies both the starting point at (P_D,P_F) = (0,0) and ending point at (P_D,P_F) = (1,1). This is the reason why most ROC curves reported in the literature, indeed, started with either P_F greater than zero or ended with P_F less than one. As a result, a normalization of A_z in Fig. 1 is necessary. This issue was overlooked and never addressed in the past. In a recent work [7], Chang investigated this problem and resolved this issue by normalizing A_z to plot ROC curves. A great deal of detailed discussions can be found in [7].

As also proved in Theorem 1 and shown in Section III-A, an ROC curve generated by $\Gamma = \{\mathbf{r}_i\}_{i=1}^N$ should be a step function with step jumps at all the data samples $\Gamma = \{\mathbf{r}_i\}_{i=1}^N$. This implies that using a uniform step size to produce an ROC curve is inappropriate, as shown in [7].

IV. Issues in Plotting 2D ROC Curves

Since P_F and P_D are actually calculated by varying the threshold variable τ used by the detector in (2), as noted in the above third issue, we can use τ instead of β to generate both values of P_D and P_F and then plot pairs of (P_D,P_F) values to construct an ROC curve. As a consequence, a true ROC curve is a step function in (32). This is probably the way that has been used in the literature to generate ROC curves, including MATLAB. This approach also gives rise to a concept of how to extend a 2D ROC curve to a 3D ROC curve developed in [7]. The key idea between a 2D ROC curve and a 3D ROC curve lies in how an ROC curve is generated. Compared to a traditional 2D ROC curve, which plots P_D versus P_F by varying $P_F = \beta$ as a variable, 3D ROC curve plots an ROC curve by varying an independent parameter τ as a threshold instead of varying $P_F = \beta$ to generate P_D as an ROC curve is commonly plotted.

Fully comprehending Theorems 1 and 2 is neither straightforward nor trivial because the τ^β in (25) is an implicit parameter, which is determined by the constraint parameter β via (8). To shed light on Theorem 1, this section illustrates the key concept behind Theorem 1 by reversing the roles of $P_F = \beta$ and $\tau = \tau^\beta$ in (8), where τ is considered as an independent variable to generate β denoted as $P_F(\tau) = \beta_\tau$. As a matter of fact, this approach paves the way for developing 3D ROC curves [7] and [34, Ch. 3].

Let $\Gamma = \{\mathbf{r}_i\}_{i=1}^N$ be the set of total data samples. There are only N values of P_F, $\{\beta_i\}_{i=1}^N$, which can be produced by $\{\Lambda(\mathbf{r}_i)\}_{i=1}^N$ to plot an 2D ROC curve. Consequently, the main issue for an RNPD is how to find $P_D(\delta^{\mathrm{NP}})$ constrained by $P_F(\delta^{\mathrm{NP}}) \leq \beta$ with $\beta \notin \{\beta_i\}_{i=1}^N$, i.e., how to calculate $P_D(\delta^{\mathrm{NP}}) =?$ when $P_F(\delta^{\mathrm{NP}}) = \beta$ with $\beta \notin \{\beta_i\}_{i=1}^N$. It is obvious that, for a continuous dataset and any given β, we can always find its corresponding τ to calculate $P_F(\delta^{\mathrm{NP}}) = \beta$ via (8) and P_D by (5). However, this will not work for a discrete set of data samples and $\beta \notin \{\beta_i\}_{i=1}^N$ because there is no corresponding data sample in $\Gamma = \{\mathbf{r}_i\}_{i=1}^N$ that can produce τ^β to yield $P_F(\delta^{\mathrm{NP}}) = \beta$ for any given β not in $\{\beta_i\}_{i=1}^N$. As a matter of fact, most works reported in the literature, including MATLAB, do not use β but rather τ to plot ROC curves via linear interpolation by RNPD in (35). Theorem 1 provides a direct approach to accomplishing this task. That is, the required approach from given $P_F(\tau) = \beta \rightarrow \tau$ is reversed to a process from given $P_F(\tau) \rightarrow \beta$.

More specifically, we assume without loss of generality that $\{\Lambda(\mathbf{r}_i)\}_{i=1}^N$ is strictly increasing sequence, i.e., $\tau_i = \Lambda(\mathbf{r}_i) < \tau_{i+1} = \Lambda(\mathbf{r}_{i+1})$ for $i = 1, 2, \ldots, N-1$. Thus, $\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) > \tau_{i+1}\} \subset_{\neq} \{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) > \tau_i\}$. For each τ, let its calculated P_F be denoted by $P_F(\tau) = \beta_\tau$. Also, for the ith data sample, $\mathbf{r}_i$, let $P_F(\tau_i) = \beta_{\tau_i} = \beta_i$ obtained by (18), i.e.,

$$P_F(\tau_i) = P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) > \tau_i\}) + \gamma(\mathbf{r}_i) P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) = \tau_i\}) = \beta_{\tau_i} = \beta_i. \quad (40)$$

For each pair of ($\Lambda(\mathbf{r}_i)$, $\Lambda(\mathbf{r}_{i+1})$) and a given threshold τ, there are three scenarios for which τ can occur, as shown in Fig. 6.

Case 1: $\tau = \tau_i = \Lambda(\mathbf{r}_i)$ with its corresponding $P_F(\tau_i) = \beta_i$.

In this case, according to (17), we declare $\delta^{\mathrm{NP}}(\mathbf{r}_{i+1}) = 1$ and $\delta^{\mathrm{NP}}(\mathbf{r}_i) = 1$ with probability $\gamma(\mathbf{r}_i)$ and $\delta^{\mathrm{NP}}(\mathbf{r}_i) = 0$ with probability $1 - \gamma(\mathbf{r}_i)$. As a result, $\gamma(\mathbf{r}_i)$ can be

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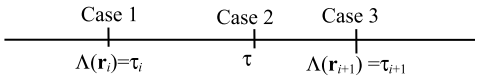


Fig. 6. Three scenarios occur for the threshold τ to generate $\mathrm{P_F}(\delta^{\mathrm{NP}}) = \beta_\tau$.

determined by (18)

$$\begin{aligned} \mathrm{P_F}(\tau_i) &= P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) > \tau_i\}) \\ &\quad + \gamma(\mathbf{r}_i) P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) = \tau_i\}) = \beta_i \\ &\Rightarrow \gamma(\mathbf{r}_i) = 1 \end{aligned} \tag{41}$$

because $P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) > \tau_i\}) < \beta_i$ and $P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) > \tau_i\}) + P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) = \tau_i\}) = \beta_i$.

Equation (41) is then further used to calculate

$$\mathrm{P_D}(\tau_i) = P_1(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) > \tau_i\}) + P_1(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) = \tau_i\}). \tag{42}$$

Case 2: $\tau_i = \Lambda(\mathbf{r}_i) < \tau < \tau_{i+1} = \Lambda(\mathbf{r}_{i+1})$ with its corresponding $\mathrm{P_F^{RNPD}}(\tau) = \beta_\tau$ for $\beta_{i+1} < \beta_\tau < \beta_i$.

In this case, $\mathbf{r}_i \in \Gamma_0$ and $\mathbf{r}_{i+1} \in \Gamma_1$. Thus, $\delta(\mathbf{r}_i) = 0$ and $\delta(\mathbf{r}_{i+1}) = 1$. Since $\beta_{i+1} < \mathrm{P_F}(\tau) = \beta_\tau < \beta_i$, from (18), we can calculate for a data sample $\mathbf{r}$ between $\mathbf{r}_i$ and $\mathbf{r}_{i+1}$

$$\begin{aligned} \mathrm{P_F^{RNPD}}(\tau) &= P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) > \tau^\beta\}) \\ &\quad + \gamma(\mathbf{r}) P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) = \tau^\beta\}) = \beta_\tau \\ \Rightarrow \gamma(\mathbf{r}) &= \frac{\beta_\tau - P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) > \tau^\beta\})}{P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) = \tau^\beta\})}. \end{aligned} \tag{43}$$

According to (25), τ^β is the smallest value in (25) such that $P_0(\Gamma_1^\beta) = P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) > \tau^\beta\}) \le \beta_\tau$. This implies that $\tau^\beta = \tau_{i+1}$ and

$$\mathrm{P_F^{RNPD}}(\tau) = \beta_\tau \Rightarrow \gamma(\mathbf{r}) = \frac{\beta_\tau - P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) > \tau_{i+1}\})}{P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) = \tau_{i+1}\})} \tag{44}$$

which derives

$$\begin{aligned} \mathrm{P_D^{RNPD}}(\beta_\tau) &= P_1(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) > \tau_{i+1}\}) \\ &\quad + \gamma(\mathbf{r}) P_1(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) = \tau_{i+1}\}). \end{aligned} \tag{45}$$

Since $P_1(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) > \tau_{i+1}\}) = \mathrm{P_D}(\beta_{i+1})$

$$\mathrm{P_D^{RNPD}}(\beta_\tau) = \mathrm{P_D}(\beta_{i+1}) + \gamma(\mathbf{r}) P_1(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) = \tau_{i+1}\}) \tag{46}$$

which is shown in Fig. 7.

However,

$$P_1(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) = \tau_i\}) = \mathrm{P_D}(\beta_i) - \mathrm{P_D}(\beta_{i+1}). \tag{47}$$

Substituting (47) into (46) yields

$$\begin{aligned} \mathrm{P_D^{RNPD}}(\beta_\tau) &= \mathrm{P_D}(\beta_{i+1}) + \gamma(\mathbf{r})(\mathrm{P_D}(\beta_i) - \mathrm{P_D}(\beta_{i+1})) \\ &= \gamma(\mathbf{r}) \mathrm{P_D}(\beta_i) + (1 - \gamma(\mathbf{r})) \mathrm{P_D}(\beta_{i+1}). \end{aligned} \tag{48}$$

Thus, for any given pair of two data samples, $\mathbf{r}_i$ and $\mathbf{r}_{i+1}$ in a discrete dataset, case 2 is the one, which can be used to linearly interpolate the missing samples between $\mathbf{r}_i$ and $\mathbf{r}_{i+1}$. In particular, the step jump at $\mathrm{P_D}(\beta_{i+1})$ in Fig. 6

$$P_1(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) = \tau_{i+1}\}). \tag{49}$$

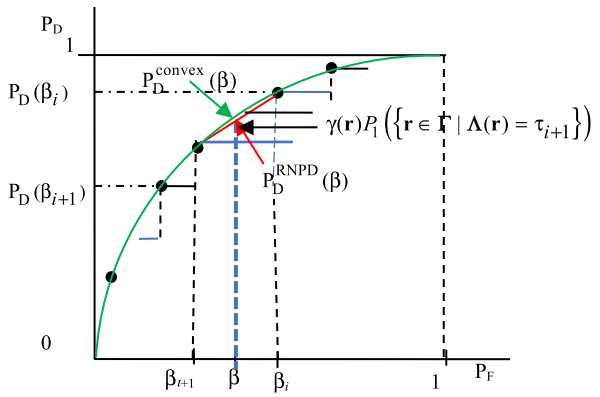


Fig. 7. Step jump at β_i with $\beta_{i+1} < \beta < \beta_i$.

Case 3: $\tau = \tau_{i+1} = \Lambda(\mathbf{r}_{i+1})$ with its corresponding $\mathrm{P_F}(\tau_{i+1}) = \beta_{i+1}$.

In this case, according to (17), we declare $\delta^{\mathrm{NP}}(\mathbf{r}_i) = 0$ and $\delta^{\mathrm{NP}}(\mathbf{r}_{i+1}) = 1$ with probability $\gamma(\mathbf{r}_{i+1})$ and $\delta^{\mathrm{NP}}(\mathbf{r}_{i+1}) = 0$ with probability $1 - \gamma(\mathbf{r}_{i+1})$. Accordingly, $\gamma(\mathbf{r}_{i+1})$ can be determined by

$$\begin{aligned} \mathrm{P_F}(\tau_{i+1}) &= P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) > \tau_{i+1}\}) \\ &\quad + \gamma(\mathbf{r}_{i+1}) P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) = \tau_{i+1}\}) = \beta_{i+1} \\ &\Rightarrow \gamma(\mathbf{r}_{i+1}) = 1 \end{aligned} \tag{50}$$

which can be further used to calculate

$$\mathrm{P_D}(\tau_{i+1}) = P_1(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) \ge \tau_{i+1}\}). \tag{51}$$

Combining all three cases actually yields a general NP randomized detector given by (22), which makes deterministic detections on the data samples, $\Gamma = \{\mathbf{r}_i\}_{i=1}^N$, and random detections on unavailable data samples $\mathbf{r} \notin \Gamma = \{\mathbf{r}_i\}_{i=1}^N$.

Using (49), we can further calculate $\mathrm{AUC}_{(\mathrm{D},\tau)}$ and $\mathrm{AUC}(\mathrm{F},\tau)$ in the following theorem.

Theorem 3:

$$\begin{aligned} \mathrm{AUC}_{\Gamma_1}^{\mathrm{true}}(\mathrm{D},\tau) &= \mathrm{AUC}_{\Gamma_1}^{\mathrm{DNPD}}(\mathrm{D},\tau) \\ &= \sum_{i=1}^{N-1} \mathrm{P_D}(\beta_i)(\tau_{i+1} - \tau_1), \quad \text{for (20)} \end{aligned} \tag{52}$$

$$\begin{aligned} \mathrm{AUC}_{\Gamma_1 \cup \bar{\Gamma}}^{\mathrm{true}}(\mathrm{D},\tau) &= \mathrm{AUC}_{\Gamma_1 \cup \bar{\Gamma}}^{\mathrm{DNPD}}(\mathrm{D},\tau) \\ &= \sum_{i=1}^{N-1} \mathrm{P_D}(\beta_{i+1})(\tau_{i+1} - \tau_1), \quad \text{for (21)} \end{aligned} \tag{53}$$

$$\begin{aligned} \mathrm{AUC}_{\Gamma_1}^{\mathrm{true}}(\mathrm{F},\tau) &= \mathrm{AUC}_{\Gamma_1}^{\mathrm{DNPD}}(\mathrm{F},\tau) \\ &= \sum_{i=1}^{N-1} \mathrm{P_F}(\tau_i)(\tau_{i+1} - \tau_1), \quad \text{for (20)} \end{aligned} \tag{54}$$

$$\begin{aligned} \mathrm{AUC}_{\Gamma_1 \cup \bar{\Gamma}}^{\mathrm{true}}(\mathrm{F},\tau) &= \mathrm{AUC}_{\Gamma_1 \cup \bar{\Gamma}}^{\mathrm{DNPD}}(\mathrm{F},\tau) \\ &= \sum_{i=1}^{N-1} \mathrm{P_F}(\tau_{i+1})(\tau_{i+1} - \tau_1), \quad \text{for (21)}. \end{aligned} \tag{55}$$

That is,

$$\mathrm{AUC}_{\Gamma_1}^{\mathrm{DNPD}}(\mathrm{D},\tau) < \mathrm{AUC}_{\Gamma_1 \cup_\gamma \bar{\Gamma}}^{\mathrm{DNPD}}(\mathrm{D},\tau) < \mathrm{AUC}_{\Gamma_1 \cup \bar{\Gamma}}^{\mathrm{DNPD}}(\mathrm{D},\tau)$$
$$\mathrm{AUC}_{\Gamma_1}^{\mathrm{DNPD}}(\mathrm{F},\tau) < \mathrm{AUC}_{\Gamma_1 \cup_\gamma \bar{\Gamma}}^{\mathrm{DNPD}}(\mathrm{F},\tau) < \mathrm{AUC}_{\Gamma_1 \cup \bar{\Gamma}}^{\mathrm{DNPD}}(\mathrm{F},\tau).$$

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Proof: For each data sample $\mathbf{r}_{i+1}$ with $0 \le i \le N-1$, its $\mathrm{P_F}(\Lambda(\mathbf{r}_{i+1}) = \tau_i) = \beta_i$ and $\mathrm{P_D}(\beta_i)$ are given by

$$\mathrm{P_D}(\beta_i) = \mathrm{P_D}(\beta_{i+1}) + P_1(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) = \tau_i\}) \tag{56}$$

where

$$\mathrm{P_D}(\beta_{i+1}) = P_1(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) \ge \tau_{i+1}\}). \tag{57}$$

Since no data samples can produce the $\mathrm{P_F}\beta$ between $\mathrm{P_F}(\Lambda(\mathbf{r}_i) = \tau_i) = \beta_i$ and $\mathrm{P_F}(\Lambda(\mathbf{r}_{i+1}) = \tau_{i+1}) = \beta_{i+1}$, the $\mathrm{P_D}$ values between $\mathrm{P_D}(\delta^{\mathrm{NP}}(\mathbf{r}_i))$ at β_i and $\mathrm{P_D}(\delta^{\mathrm{NP}}(\mathbf{r}_{i+1}))$ at β_{i+1} are not obtainable. In this case, linearly interpolating a 2D ROC curve is a common practice in hyperspectral detection, including MATLAB software. That is, many reports that use MATLAB to plot 2D ROC curves have used $\mathrm{P_D}(\delta^{\mathrm{NP}}(\mathbf{r}_i))$ at β_i and $\mathrm{P_D}(\delta^{\mathrm{NP}}(\mathbf{r}_{i+1}))$ at β_{i+1} as two endpoints to linearly interpolate $\mathrm{P_D}(\delta^{\mathrm{NP}}(\mathbf{r}))$ for $\mathrm{P_F}$ β in the range of $\beta_{i+1} < \beta < \beta_i$, with the randomized parameter $0 \le \gamma(\mathbf{r}) \le 1$ specified by (34) so that $\mathrm{P_D}(\beta)$ can be calculated by

$$\mathrm{P_D}(\beta) = (1 - \gamma(\mathbf{r}))\mathrm{P_D}(\beta_{i+1})) + \gamma(\mathbf{r})\mathrm{P_D}(\beta_i)) \tag{58}$$

which is shown by the red line in Fig. 6

$$\mathrm{AUC}^{\mathrm{RNPD}}_{\Gamma_1 \cup \gamma \bar{\Gamma}}(\mathrm{D}, \tau) = \sum_{i=1}^{N-1} \mathrm{P_D}(\beta)(\tau_{i+1} - \tau_1), \quad \text{for (22)} \tag{59}$$

$$\mathrm{AUC}^{\mathrm{RNPD}}_{\Gamma_1 \cup \gamma \bar{\Gamma}}(\mathrm{F}, \tau) = \sum_{i=1}^{N-1} \mathrm{P_F}(\tau)(\tau_{i+1} - \tau_1), \quad \text{for (22)}. \tag{60}$$

■

For a discrete set of hyperspectral image pixels, such a data sample in (58) does not exist. Thus, technically speaking, using (58) to linearly interpolate an ROC curve is only appropriate for a discrete set of data samples. For a continuous dataset, the interpolation should be carried out by a convex-cap function, not a linear function shown by the green line in Fig. 7.

It is worth noting that there is a key and significant difference between the unavailability of data samples and data samples. For no data samples available, such as hyperspectral images, (51) is not valid. By contrast, for unavailable data samples, such as continuous signals commonly used in communications and signal processing, (58) can be used as upsampling but generally cannot be accurate since the interpolation should be convex-cap shown by the green line in Fig. 7. Nevertheless, when a discrete set of data samples, the ROC curve of an RNPD can be obtained by the red-highlighted by a linear interpolation function via (58). As the number of data samples is increased, the gap difference between the interpolated green and red lines will approach zero, as shown in Fig. 7.

V. NP Detection of a Binary Communication Channel

This section presents a discrete example to illustrate the idea of Theorems 1 and 2 and discussions in Section IV. It is a binary communication channel discussed in [1] and shown in Fig. 8 which has two hypotheses specified by

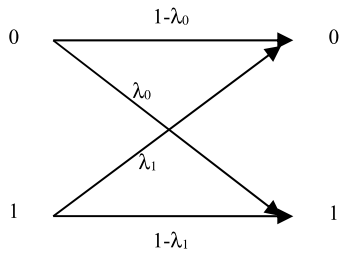


Fig. 8. Binary communication channel.

$$\begin{aligned} &H_0 : 0\text{transmitted} \sim p_0(\mathbf{r}) \\ &\text{versus} \\ &H_1 : 1\text{transmitted} \sim p_1(\mathbf{r}) \end{aligned} \tag{61}$$

where $\Gamma = \{0, 1\}$ and probability distributions under H_0 and H_1 are given by $p(\mathbf{r} = 1|H_0) = p_0(\mathbf{r} = 1) = \lambda_0$ and $p(\mathbf{r} = 0|H_1) = p_1(\mathbf{r} = 0) = \lambda_1$.

By interpreting a discrete dataset, $\Gamma = \{0, 1\}$ with the two data samples $\mathbf{r}_i = \mathbf{r}_0 = 0$ and $\mathbf{r}_i = \mathbf{r}_1 = 1$, the general RNPD derived in Section IV-A can be well-explained by the example of a binary communication channel presented in the following treatment where: $\tau_0 = \Lambda(\mathbf{r}_0 = 0) < \tau_1 = \Lambda(\mathbf{r}_1 = 1)$.

The RNPD obtained from (61) is now given by

$$\delta^{\mathrm{RNPD}}(\mathbf{r}) = \begin{cases} 1; & \text{if } \Lambda(\mathbf{r}) > \tau \\ 1, & \text{with probability } \gamma(\mathbf{r}) \text{ if } \Lambda(\mathbf{r}) = \tau \\ 0; & \text{if} \Lambda(\mathbf{r}) < \tau \end{cases} \tag{62}$$

where $\Lambda(\mathbf{r}) = ((p_1(\mathbf{r}))/(p_0(\mathbf{r})))$ is LRT given by

$$\Lambda(\mathbf{r} = 0) \equiv \Lambda(\mathbf{r}_0) = \frac{p_1(\mathbf{r} = 0)}{p_0(\mathbf{r} = 0)} = \frac{\lambda_1}{1 - \lambda_0} \tag{63}$$

$$\Lambda(\mathbf{r} = 1) \equiv \Lambda(\mathbf{r}_1) = \frac{p_1(\mathbf{r} = 1)}{p_0(\mathbf{r} = 1)} = \frac{1 - \lambda_1}{\lambda_0}. \tag{64}$$

If $\lambda_0 + \lambda_1 < 1$, then $\Lambda(\mathbf{r}_0) < \Lambda(\mathbf{r}_1)$. In this case, Fig. 6 becomes Fig. 8 where there are five cases.

Case 1: $\tau < \Lambda(\mathbf{r}_0) = \frac{\lambda_1}{1-\lambda_0}$ with its corresponding $\mathrm{P_F}(\tau) = \beta_\tau$.

In this case, according to (61), $\Gamma_0 = \bar{\Gamma} = \emptyset$ and $\Gamma_1 = \Gamma = \{0, 1\}$. Thus, we declare $\delta^{\mathrm{NP}}(\mathbf{r}) = 1$ for all $\mathbf{r} = 0, 1$ with probability $\gamma(\mathbf{r}) = 0$. As a result, $\mathrm{P_F}(\tau) = \beta_\tau = 1$ and $\mathrm{P_D}(\tau) = 1$.

Now, for $\lambda_0 = \mathrm{P_F}(\tau_0) = P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) > \tau_0\}) < \beta_\tau < 1$, we need to linearly interpolate the unavailable data sample $\mathbf{r}$ whose $\mathrm{P_F}$ value is greater than β_1. Then, $\gamma(\mathbf{r})$ can be determined by

$$\begin{aligned} \mathrm{P_F}(\beta_\tau) &= P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) > \tau^{\beta_\tau}\}) \\ &\quad + \gamma(\mathbf{r}) P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) = \tau^{\beta_\tau}\}) = \beta_\tau \\ &\Rightarrow \tau^{\beta_\tau} = \tau_0 \end{aligned} \tag{65}$$

$$\begin{aligned} \mathrm{P_F}(\beta_\tau) &= P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) > \tau_0\}) \\ &\quad + \gamma(\mathbf{r}) P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) = \tau_0\}) = \beta \\ &\Rightarrow \lambda_0 + \gamma(\mathbf{r})(1 - \lambda_0) = \beta \Rightarrow \gamma(\mathbf{r}) = \frac{\beta - \lambda_0}{1 - \lambda_0}. \end{aligned} \tag{66}$$

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Using (66), we can further calculate

$$\begin{aligned} \mathrm{P_D}(\beta_\tau) &= P_1(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) > \tau_0\}) \\ &\quad + \gamma(\mathbf{r}) P_1(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) = \tau_0\}) \\ &= (1-\lambda_1) + \frac{\beta_\tau - \lambda_0}{1-\lambda_0}\lambda_1. \end{aligned} \tag{67}$$

Case 2: $\tau_0 = \Lambda(\mathbf{r}_0) = (\lambda_1/(1-\lambda_0))$ with its corresponding $\mathrm{P_F}(\tau_0) = \beta_{\tau_0} = \beta_0 \Rightarrow \mathrm{P_0}(\Gamma_1) = \lambda_0$.

In this case, according to (61), $\Gamma_0 = \emptyset$, $\bar{\Gamma} = \{0\}$, and $\Gamma_1 = \{1\}$. Thus, we declare $\delta^{\mathrm{NP}}(\mathbf{r}_1) = \delta^{\mathrm{NP}}(\mathbf{r} = 1) = 1$ and $\delta^{\mathrm{NP}}(\mathbf{r}_0) = \delta^{\mathrm{NP}}(\mathbf{r} = 0) = 1$ with probability $\gamma(\mathbf{r}_0)$ and $\delta^{\mathrm{NP}}(\mathbf{r}_0) = 0$ with probability $1 - \gamma(\mathbf{r}_0)$. Accordingly, from (51), $\gamma(\mathbf{r}_0) = ((\beta - \lambda_0)/(1-\lambda_0)) = 0$ by letting $\beta = \beta_0$ with $\mathrm{P_F}(\tau_0) = \beta_0$. In addition, $\mathrm{P_D}(\tau_0)$ can be determined by letting $\beta_\tau = \beta_0$

$$\mathrm{P_D}(\tau_0) = P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) > \tau_0\}) = p_0(\mathbf{r} = 1) = 1 - \lambda_1. \tag{68}$$

Case 3: $\tau_0 = \Lambda(\mathbf{r}_0) < \tau < \tau_1 = \Lambda(\mathbf{r}_1)$ with its corresponding $\mathrm{P_F}(\tau) = \beta_\tau$.

In this case, $\Gamma_0 = \{0\}$, $\Gamma_1 = \{1\}$, and $\bar{\Gamma} = \emptyset$. Since $\bar{\Gamma} = \{\mathbf{r} \in \Gamma = \{0,1\} | \Lambda(\mathbf{r}) = \tau\} = \emptyset$, this implies that $\gamma(\mathbf{r}_0) = 0$ and

$$\mathrm{P_F}(\tau) = P_0(\Gamma_1) = \beta_0 = \lambda_0 \tag{69}$$

which derives

$$\mathrm{P_D}(\tau) = P_1(\Gamma_1) = 1 - \lambda_1. \tag{70}$$

According to (69) and (70), $\mathrm{P_F}(\delta^{\mathrm{NP}})$ and $\mathrm{P_D}(\delta^{\mathrm{NP}})$ is constant for $\tau_0 = \Lambda(\mathbf{r}_0) < \tau < \tau_1 = \Lambda(\mathbf{r}_1)$ because all the values between two discrete values, $\Lambda(\mathbf{r}_0)$ and $\Lambda(\mathbf{r}_1)$, resulting from discrete data samples are missing, and thus, $\bar{\Gamma} = \emptyset$. This indicates that no data samples in $\Gamma = \{0,1\}$ can yield $\mathrm{P_F}\beta$ in the range of $\lambda_0 < \beta < \beta_1$, which also explains why an ROC curve is a step function with step jumps occurred at each data sample. In other words, the case of $\tau_0 = \Lambda(\mathbf{r}_0) < \tau < \tau_1 = \Lambda(\mathbf{r}_1)$ never occurs because no data samples in $\Gamma = \{0,1\}$ can produce $\beta_0 < \mathrm{P_F}(\tau) = P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) > \tau\}) = \beta_\tau < \beta_1$. Under such a circumstance, we need RNPD to linearly interpolate the unavailable data samples, where $\gamma(\mathbf{r}_1)$ can be determined by

$$\begin{aligned} \mathrm{P_F}(\tau) &= P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) > \tau^\beta\}) \\ &\quad + \gamma(\mathbf{r}_1) P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) = \tau^\beta\}) = \beta \\ &\Rightarrow \tau^\beta = \tau_0 \end{aligned} \tag{71}$$

which implies that

$$\begin{aligned} \mathrm{P_F}(\tau) &= P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) > \tau^\beta\}) \\ &\quad + \gamma(\mathbf{r}_0) P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) = \tau^\beta\}) = \beta \\ \Rightarrow \gamma(\mathbf{r}_0) &= \frac{\beta - p_0(\mathbf{r}=1)}{p_0(\mathbf{r}=0)} = \frac{\beta - \lambda_0}{1-\lambda_0}. \end{aligned} \tag{72}$$

Using (72) can further calculate

$$\mathrm{P_D}(\tau_0) = (1-\lambda_1) + \gamma(\mathbf{r}_0)\lambda_1 = (1-\lambda_1) + \frac{\beta_0 - \lambda_0}{1-\lambda_0}\lambda_1. \tag{73}$$

Case 4: $\tau = \tau_1 = \Lambda(\mathbf{r}_1)$ with its corresponding $\mathrm{P_F}(\tau_1) = \beta_{\tau_1} = \beta_1 \Rightarrow P_0(\Gamma_1) = 0$.

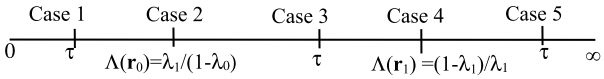


Fig. 9. Five scenarios occur for the threshold τ to generate $\mathrm{P_F}(\delta^{\mathrm{NP}}) = \beta_\tau$.

In this case, according to (61), $\Gamma_0 = \{0\}$, $\bar{\Gamma} = \{1\}$, and $\Gamma_1 = \emptyset$. Thus, we declare $\delta^{\mathrm{NP}}(\mathbf{r}_0) = 0$ and $\delta^{\mathrm{NP}}(\mathbf{r}_1) = 1$ with probability $\gamma(\mathbf{r}_1)$ and $\delta^{\mathrm{NP}}(\mathbf{r}_1) = 0$ with probability $1 - \gamma(\mathbf{r}_1)$. Accordingly, $\gamma(\mathbf{r}_1)$ can be determined by

$$\begin{aligned} \mathrm{P_F}(\tau_1) &= P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) > \tau_1\}) \\ &\quad + \gamma(\mathbf{r}_1) P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) = \tau_1\}) = \beta_1 \\ &\Rightarrow \gamma(\mathbf{r}_1) = \frac{\beta_1 - P_0(\Gamma_1)}{P_0(\in \bar{\Gamma})} = \frac{\beta_1}{\lambda_0} \end{aligned} \tag{74}$$

which can be further used to calculate

$$\mathrm{P_D}(\tau_1) = \frac{\beta_1}{\lambda_0}(1-\lambda_1). \tag{75}$$

Case 5: $\Lambda(\mathbf{r}_1) = ((1-\lambda_1)/\lambda_0) < \tau$ with its corresponding

$$\begin{aligned} &\mathrm{P_F}(\tau_1) = \beta_1 \\ &\Rightarrow \mathrm{P_F}(\tau_1) = P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) > \tau_1\}) = 0. \end{aligned} \tag{76}$$

In this case, according to (61), $\Gamma_0 = \Gamma = \{0,1\}$ and $\Gamma_1 = \bar{\Gamma} = \emptyset$. Thus, we declare $\delta^{\mathrm{NP}}(\mathbf{r}) = 0$ for all $\mathbf{r} = 0, 1$ with probability $\gamma(\mathbf{r}) = 0$. As a result, $\mathrm{P_D}(\tau) = 0$ and $\mathrm{P_F}(\tau) = \beta_\tau = 0$.

Now, for $0 < \beta_\tau < \beta_1 = \mathrm{P_F}(\tau_1) = \lambda_0$ given by (76), we need to linearly interpolate the unavailable data samples $\mathbf{r}$ whose $\mathrm{P_F}$ value is less than β_1. Then, $\gamma(\mathbf{r})$ can be determined by

$$\begin{aligned} \mathrm{P_F}(\beta_\tau) &= P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) > \tau^{\beta_\tau}\}) \\ &\quad + \gamma(\mathbf{r}) P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) = \tau^{\beta_\tau}\}) = \beta_\tau \\ &\Rightarrow \tau^{\beta_\tau} = \tau_1 \end{aligned} \tag{77}$$

which yields

$$\begin{aligned} \mathrm{P_F}(\beta_\tau) &= P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) > \tau_1\}) \\ &\quad + \gamma(\mathbf{r}) P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) = \tau_1\}) = \beta. \end{aligned} \tag{78}$$

Since $P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) > \tau_1\}) = P_0(\emptyset) = 0$

$$\begin{aligned} &\mathrm{P_F}(\beta_\tau) = \gamma(\mathbf{r}) P_0(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) = \tau_1\}) = \beta_1 \\ &\Rightarrow \gamma(\mathbf{r}) = \frac{\beta_1}{\lambda_0}. \end{aligned} \tag{79}$$

Using (79), we can further calculate

$$\begin{aligned} \mathrm{P_D}(\beta_\tau) &= \gamma(\mathbf{r}) P_1(\{\mathbf{r} \in \Gamma | \Lambda(\mathbf{r}) = \tau_1\}) \\ &= \frac{\beta_\tau}{\lambda_0}(1-\lambda_1). \end{aligned} \tag{80}$$

Fig. 10 shows $\mathrm{P_D}$ and $\mathrm{P_F}$ calculated for the above five cases.

Fig. 11 plots the ROC curve of $\mathrm{P_D}$ versus $\mathrm{P_F}$ for the binary channel specified by (56) with $\mathrm{P_F}(\tau_0) = p_0(\mathbf{r} = 1) = \lambda_0$ where the randomized detectors for the unavailable data samples are obtained by linear interpolation highlighted by red lines, but the green line shows its continuous convex-cap interpolation. Nevertheless, both are, indeed, convex-cap functions.

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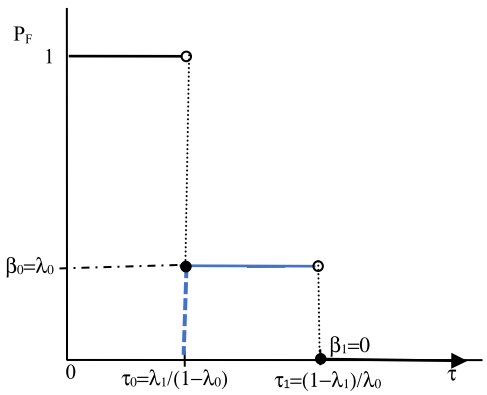


Fig. 10. Relationship between $P_F = \beta$ and the threshold τ.

Thus, according to Figs. 9 and 10 and the five scenarios discussed above, we summarize as follows.

For τ in the above five scenarios

$$P_0(\Lambda(\mathbf{r}) > \tau) = \begin{cases} 1; & \text{case 1: } \tau < \dfrac{\lambda_1}{1-\lambda_0} \\ \lambda_0; & \text{case 2: } \tau = \dfrac{\lambda_1}{1-\lambda_0} \\ \lambda_0; & \text{case 3: } \dfrac{\lambda_1}{1-\lambda_0} < \tau < \dfrac{1-\lambda_1}{\lambda_0} \\ 0; & \text{case 4: } \tau = \dfrac{1-\lambda_1}{\lambda_0} \\ 0; & \text{case 5: if } \tau > \dfrac{1-\lambda_1}{\lambda_0} \end{cases} \tag{81}$$

which can be combined into three scenarios

$$P_0(\Lambda(\mathbf{r}) > \tau) = \begin{cases} 1; & \text{case 1: } \tau < \dfrac{\lambda_1}{1-\lambda_0} \\ \lambda_0; & \text{cases 2-3: } \dfrac{\lambda_1}{1-\lambda_0} \le \tau < \dfrac{1-\lambda_1}{\lambda_0} \\ 0; & \text{cases 4-5: } \dfrac{1-\lambda_1}{\lambda_0} \le \tau \end{cases} \tag{82}$$

with τ^β in (25) given by

$$\tau^\beta = \begin{cases} \dfrac{1-\lambda_1}{\lambda_0} = \Lambda(\mathbf{r}=0); & \text{if } 0 \le \beta < \lambda_0 \\ \dfrac{\lambda_1}{1-\lambda_0} = \Lambda(\mathbf{r}=0); & \text{if } \lambda_0 \le \beta < 1 \\ 0; & \text{if } \beta = 0 \end{cases} \tag{83}$$

and $\gamma(\mathbf{r})$ given by

$$\gamma(\mathbf{r}) = \begin{cases} \dfrac{\beta}{\lambda_0}; & \text{if } 0 \le \beta < \lambda_0 \\ \dfrac{\beta-\lambda_0}{1-\lambda_0}; & \text{if } \lambda_0 \le \beta < 1 \\ \text{arbitrary}; & \text{if } \beta = 1. \end{cases} \tag{84}$$

By (82)–(84), the RNPD $\delta^{\text{NP}}(\mathbf{r})$ can be obtained by

$$\delta_\beta^{\text{NP}}(\mathbf{r}) = \begin{cases} \dfrac{\beta}{\lambda_0}; & \text{if } \mathbf{r} = 1 \\ 0; & \text{if } \mathbf{r} = 0 \end{cases} \tag{85}$$

for $0 \le \beta < \lambda_0$ and

$$\delta_\beta^{\text{NP}}(\mathbf{r}) = \begin{cases} 1; & \text{if } \mathbf{r} = 1 \\ \dfrac{\beta-\lambda_0}{1-\lambda_0}; & \text{if } \mathbf{r} = 0 \end{cases} \tag{86}$$

for $\lambda_0 \le \beta \le 1$.

$P_D(\delta_\beta^{\text{NP}})$ with a specified $P_F = \beta$ is now given by

$$P_D(\delta_\beta^{\text{NP}}) = \begin{cases} \beta\dfrac{1-\lambda_1}{\lambda_0}, & \text{if } 0 \le \beta < \lambda_0 \\ (1-\lambda_1) + \lambda_1\dfrac{\beta-\lambda_0}{1-\lambda_0}; & \text{if } \lambda_0 \le \beta \le 1. \end{cases} \tag{87}$$

VI. AD as an Anomaly-BKG Detection Problem

The binary communication channel example presented in Section V serves as a base for formulating the anomaly-BKG AD problem described in this section. It uses H_0 that specifies "0" transmitted in (53) to interpret BKG, while using H_1 that specifies "1" transmitted" in (53) to detect an anomaly. Accordingly, what is derived for the binary communication channel in Section IV-B is readily applied to the anomaly-BKG detection problem formulated as follows:

$$\begin{aligned} &H_0 : \mathbf{r} \approx \text{BKG} \\ &\text{versus} \\ &H_1 : \mathbf{r} \approx \text{anomlay} + \text{BKG} \end{aligned} \tag{88}$$

where BKG and anomaly are specified by two hypotheses H_0 and H_1, called "*null hypothesis,*" respectively.

To address the two-class anomaly-BKG detection problem specified by (88), a confusion matrix developed in [34] instead of P_D and P_F can be used for this purpose with the following definitions.

1) n_{AD} is the number of detected anomalous samples.
2) n_{CAD} is the number of correctly detected anomalous samples.
3) n_{BD} is the number of BKG samples detected as BKG samples.
4) n_{CBD} is the number of correctly detected BKG samples.
5) n_{FA} is the number of BKG samples falsely detected as anomalous samples.
6) n_{FB} is the number of anomalous samples falsely detected as BKG samples, i.e., anomalous samples missed by detection.
7) n_{A} is the number of anomalous samples.
8) n_{B} is the number of BKG samples.
9) N is the total number of data samples $= n_{\text{A}} + n_{\text{B}}$.

Now, with the above definitions, we can have

$$n_A = n_{\text{CAD}} + n_{\text{FB}} \tag{89}$$

$$n_B = n_{\text{CBD}} + n_{\text{FA}} \tag{90}$$

$$\text{P}(H_0) = \text{P(BKG)} = \frac{n_B}{N} \tag{91}$$

$$\text{P}(H_1) = \text{P(anomaly)} = \frac{n_A}{N} \tag{92}$$

and also define the following various detection measures that can be defined for AD.

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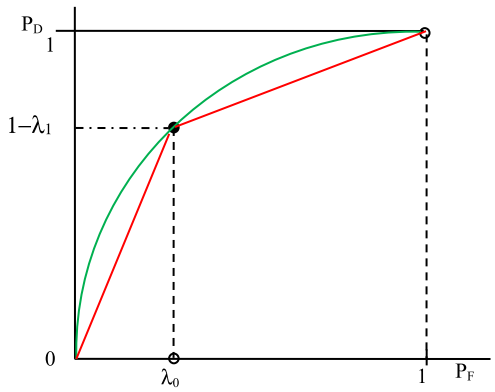


Fig. 11. Linear interpolation of the ROC curve of (P_D,P_F) by RNPD for a binary communication channel.

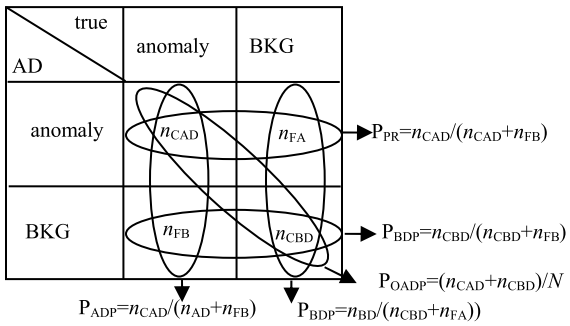


Fig. 12. Confusion matrix for a target-BKG two-class detection problem.

1) *ADP:*

$$\mathrm{P_{ADP}} = \frac{n_{\mathrm{CAD}}}{n_A} = \frac{n_{\mathrm{CAD}}}{n_{\mathrm{CAD}} + n_{\mathrm{FB}}} \tag{93}$$

which is also referred to as recall rate or sensitivity.

2) *False Anomaly Probability (FAP):*

$$\mathrm{P_{FAP}} = \frac{n_{\mathrm{FA}}}{n_{\mathrm{FA}} + n_{\mathrm{CAD}}}. \tag{94}$$

3) *BKG Detection Probability (BDP):*

$$\mathrm{P_{BDP}} = \mathrm{P_{specifity}} = \frac{n_{\mathrm{CBD}}}{n_{\mathrm{CBD}} + n_{\mathrm{FB}}} \tag{95}$$

which is also referred to as predictive value (BPV)/ rate (BPR).

4) *OADP:*

$$P_{\mathrm{OADP}} = \frac{n_{\mathrm{CAD}} + n_{\mathrm{CBD}}}{n_A + n_B}. \tag{96}$$

Fig. 12 depicts a confusion matrix for the two-TD problem described by (88).

Although a 3D ROC analysis has been developed for TD specified by (1) in [7], AD is quite different from TD in the sense that it does not require any prior target knowledge, as noted in [3] and [4]. Since anomalies are unknown targets embedded in BKG, unlike TD that is the detection of a known target in noise, AD cannot be generally described by a binary hypothesis testing problem in (1), as shown in [8].

It should be noted that the idea of using the parameter τ as a variable to generate an ROC curve provides the fundamental concept behind the 3D ROC analysis, which was first envisioned in [35] and later expanded to chemical remote sensing in [36] and [37], biometrics in [38], [39], and [40], medical imaging [41], vital sign detection [42], [43], hyperspectral image classification in [44] with more detailed discussions on applications in multi-parameter ROC analysis [45], and hyperspectral data exploitation in [34, Ch. 3]. Nevertheless, Chang [7] provided a very thorough review of 3D ROC analysis with a comprehensive analysis for hyperspectral TD.

Since BS and ADBS are crucial indicators for AD performance, a 3D ROC analysis exclusively developed for AD is a necessity. As noted in (2), $\mathrm{P_F}$ and $\mathrm{P_D}$ are actually calculated by varying the threshold variable τ used by the detector. In this case, we can use τ instead of β to generate both values of $\mathrm{P_D}$ and $\mathrm{P_F}$ and then plot pairs of ($\mathrm{P_D}$,$\mathrm{P_F}$) to construct an ROC curve. As a consequence, an ROC curve is a step function as noted in the above second issue. This is probably the way that has been used in the literature to generate ROC curves, including MATLAB. This approach also gives rise to a concept of how to extend a 2D ROC curve to a 3D ROC curve developed in [7]. The key idea between a 2D ROC curve and a 3D ROC curve lies in how an ROC curve is generated. Compared to a traditional 2D ROC curve that plots $\mathrm{P_D}$ versus $\mathrm{P_F}$ by varying $\mathrm{P_F}(\delta^{\mathrm{NP}}) = \beta$ as a variable, the 3D ROC curve plots an ROC curve by varying an independent parameter τ as a threshold instead of varying $\mathrm{P_F}(\delta^{\mathrm{NP}}) = \beta$ to generate $\mathrm{P_D}$ as an ROC curve is commonly plotted.

By virtue of a 3D ROC curve, three 2D ROC curves can be generated, each of which can be used for a particular task, that is, the 2D ROC curve of ($\mathrm{P_D}$,$\mathrm{P_F}$) used to evaluate the effectiveness of a detector, the 2D ROC curve of ($\mathrm{P_D}$,τ) used to dictate anomaly detectability of a detector, and the 2D ROC curve of ($\mathrm{P_F}$,τ) used to quantify the degree of BKG suppressibility. According to Fig. 11, various detection measures are readily defined as follows.

A. Anomaly Detection Probability

The AUC value of 2D ROC curves of ($\mathrm{P_D}$,τ) is commonly used to measure the effectiveness of a detector. It can be also used to evaluate how effective an anomaly detector is. That is, the higher the value of $\mathrm{AUC_{(D,\tau)}}$, the better the anomaly detector. In this case, we can define a measure, called ADP of a detector as $\mathrm{AUC_{ADP}}$ by

$$0 \le \mathrm{AUC_{ADP}} = \mathrm{AUC_{(D,\tau)}} \le 1 \tag{97}$$

to quantitatively calculate the anomaly detectability of a detector.

B. Background Detection Probability

In TD, $\mathrm{P_F}$ is the false alarm probability, and 1-$\mathrm{P_F}$ is the detection of noise. Thus, using 1-$\mathrm{P_F}$ to evaluate noise detection performance does not make sense. However, this is not true from a perspective medical diagnosis where 1-$\mathrm{P_F}$ is defined as specificity probability or true negative probability. Also, from the binary hypothesis formulation for AD specified by (88), $\mathrm{P_F}$ represents the probability of BKG data samples falsely detected as anomalies. With this interpretation, 1-$\mathrm{P_F}$ is, indeed,

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TABLE I
RELATIONSHIP EQUIVALENCE BETWEEN CLASSIC DETECTION METRICS AND THEIR CORRESPONDING 3D ROC AUC DETECTION MEASURES

3D ROC AUC measures / classic detection metrics	$AUC_{(D,F)}$	$AUC_{(F,\tau)}$	$AUC_{ADP}=AUC_{(D,\tau)}$	$AUC_{BDP}=1-AUC_{(F,\tau)}$	AUC_{JAD}	AUC_{JBS}	AUC_{ADBS}	AUC_{SNPR}	AUC_{OADP}
(P_D,P_F)	√								
P_D (30)/(92)			√						
P_F (28)/(93)		√		√					
recall/sensitivity (92)			√						
specificity (95)				√					
OA (100)							√		
Youden's *J* statistic (102)							√ (103)		
IOU (105)								√ (106)	

the probability of correctly detecting BKG samples. Thus, by virtue of the AUC value of 2D ROC curves of (P_F,τ), $AUC_{(F,\tau)}$, we can define BDP, AUC_{BDP}, as a counterpart of ADP in (95) by

$$0 \le AUC_{BDP} = 1 - AUC_{(F,\tau)} \le 1. \tag{98}$$

C. Joint Anomaly Detectability With Background Suppressibility

A 2D ROC curve of (P_D,P_F) has been widely used to measure the effectiveness of a detector [1]. As pointed out in Sections II and III, these P_D and P_F are actually calculated by the same threshold τ for their joint performance without accounting in ADP and BDP separately. In order to factor $AUC_{(F,\tau)}$ into $AUC_{(D,\tau)}$, we need to define the joint anomaly detectability and BKG suppressibility (ADBS) of a detector.

1) *AUC_{JAD}:* Joint AD of a detector defined by

$$0 \le AUC_{JAD} = AUC_{(D,F)} + AUC_{ADP} \le 2. \tag{99}$$

2) *AUC_{JBS}:* Joint BS of detector defined by

$$0 \le AUC_{JBS} = AUC_{(D,F)} + AUC_{BDP} \le 2. \tag{100}$$

D. AD in BKG (AD-BS)

The first ADBS measure is to consider AD and BS working separately with different values of the threshold t as an independent parameter. Since P_{FA} is caused by the probability of misdetecting BKG as an anomaly, this error probability should be subtracted from P_{AD}. To take care of this effect, a measure, factoring P_{FA} into P_{AD}, to be called ADBS, can be defined as AUC_{ADBS} by

$$0 \le AUC_{ADBS} = AUC_{ADP} + AUC_{BDP} = AUC_{OA} \le 2 \tag{101}$$

which is exactly the same as OA commonly used in binary classification. In medical applications, we can interpret AUC_{ADP} and AUC_{BDP} as abnormal and normal cases, respectively. Then, AUC_{ADP} and AUC_{BDP} are known as sensitivity and specificity widely used in medical diagnosis

$$AUC_{MD} = \text{sensitivity} + \text{specificity} = AUC_{ADBS} \le 2. \tag{102}$$

Interestingly, Youden's J statistic is a single statistic that captures the performance of a dichotomous diagnostic test,[1] which is defined as

$$\begin{aligned} -1 \le J &= \text{sensitivity} + \text{specificity} - 1 \le 1 \\ \Rightarrow J &= AUC_{MD} - 1 = AUC_{OA} - 1 \end{aligned} \tag{103}$$

or alternatively

$$\begin{aligned} J &= \text{sensitivity} - (1 - \text{specificity}) \\ \Rightarrow J &= AUC_{(D,\tau)} - AUC_{(F,\tau)} = AUC_{ADBS} - 1. \end{aligned} \tag{104}$$

According to (103) and (104), Youden's J statistic is essentially the same as AUC_{ADBS}.

E. Signal-to-Noise Probability Ratio

The most effective detection measure comes from a similar idea to a criterion widely used in communications/signal processing, called SNR, and can be defined as SNPR by

$$0 \le AUC_{SNPR} = \frac{AUC_{(D,\tau)}}{AUC_{(F,\tau)}} \tag{105}$$

where P_F can be assumed to be caused by noise.

Interestingly, a commonly used metric for object detection, called IOU in computer vision,[2] also known as the Tanimoto index (TI) [46] or the Jaccard index, is defined as

$$\begin{aligned} IOU &= \frac{|A \cap B|}{|A \cup B|} = \frac{n_{AP}}{n_{AP} + n_{FA} + n_{FB}} = \frac{n_{AP}}{n_A + n_{FA}} \\ \Rightarrow IOU &= \frac{n_{AP}/n_A}{1 + n_{FA}/n_A} = \frac{P_{ADP}}{1 + n_{FA}/n_A} \end{aligned} \tag{106}$$

where $|A|$ is defined as the size of set A. If n_A can be approximated by $n_{AD}+n_{FA}$, then $n_{FA}/n_A \approx P_{FAP}$. In this case, P_{ADP} and P_{FAP} can be interpreted as $AUC_{(D,\tau)}$ and $AUC_{(F,\tau)}$ to yield its corresponding AUC measure given by

$$AUC_{IOU} = \frac{AUC_{(D,\tau)}}{1 + AUC_{(F,\tau)}}. \tag{107}$$

Comparing (107) to (105) the only difference is in their denominators where there is extra "1" in (107). Accordingly, AUC_{IOU} in (107) can be also considered a variant of AUC_{SNPR}.

[1] https://en.wikipedia.org/wiki/Youden%27s_J_statistic

[2] https://pyimagesearch.com/2016/11/07/intersection-over-union-iou-for-object-detection/

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F. Overall Anomaly Detection Probability

The last detection measure combines ADP and BDP along with $\mathrm{AUC}_{(\mathrm{D,F})}$ to derive the overall detection probability (ODP), $\mathrm{AUC}_{\mathrm{ODP}}$, of a detector defined by

$$0 \leq \mathrm{AUC}_{\mathrm{OADP}} = \mathrm{AUC}_{(\mathrm{D,F})} + \mathrm{AUC}_{\mathrm{ADP}} + \mathrm{AUC}_{\mathrm{BDP}} \leq 3. \quad (108)$$

Table I provides relationship equivalence between commonly used detection metrics and *a priori* 3D ROC curve-derived AUC detection measures where a check symbol "$\surd$" indicates a relationship between a detection metric specified by its equation in parenthesis and its corresponding 3D ROC AUC measures. In signal detection, the ROC curve is designed to evaluate the performance of a detector in noise. Thus, according to Table I, some AUC detection measures do not find their corresponding counterparts in classic detection metrics, such as $\mathrm{AUC}_{\mathrm{JAD}}$, $\mathrm{AUC}_{\mathrm{JBS}}$, and $\mathrm{AUC}_{\mathrm{OADP}}$. Interestingly, the great potential of the 3D ROC curve-based approach has also been found in object detection [47], [48]. Specifically, there is a detection measure, called PR, which is used to evaluate *a posteriori* performance of a classifier based on classified data samples for binary classification. Such PR can be also applied to the AD. It can be defined as the number of correctly detected anomalies, n_{CAD} divided by the number of all correctly detected data samples, n_{CAD} plus the number of incorrectly detected data samples as false alarm data samples, and n_{FA} given by

$$\mathrm{P}_{\mathrm{PR}} = \frac{n_{\mathrm{CAD}}}{n_{\mathrm{CAD}} + n_{\mathrm{FA}}}. \quad (109)$$

In this case, P_{PR} is also referred to as AP, i.e., $\mathrm{P}_{\mathrm{AP}} = \mathrm{P}_{\mathrm{PR}}$. Thus, P_{AP} can be referred to as anomaly predictive value (APV). The difference between (109) and (93) is that (109) uses the detected samples to measure AD performance, while (93) uses ground truth to measure AD performance. Consequently, AP can be considered a posteriori detection measure as opposed to (93), which can be considered a priori *detection* measure, ADP. With this interpretation, the 3D ROC curve-derived AUC detection measures in [7] are, indeed, *a priori* detection measures. This is the reason that AP has not been used in TD and is not included in [7]. Most interestingly, using (109) can further allow us to define its corresponding AUC measure as follows:

$$\mathrm{AUC}_{\mathrm{AP}} = \frac{\mathrm{AUC}_{\mathrm{ADP}}}{\mathrm{AUC}_{\mathrm{ADP}} + (1 - \mathrm{AUC}_{\mathrm{BDP}})}. \quad (110)$$

Another interesting a posteriori detection measure that has been used for object detection but not for AD is the F_1-score to measure the accuracy of a detector. It is calculated from the precision, AP, and ADP as the harmonic mean of AP and ADP by

$$\mathrm{F}_1 = 2\frac{\mathrm{ADP} \cdot \mathrm{AP}}{\mathrm{ADP} + \mathrm{AP}} = 2\frac{n_{\mathrm{CAD}}}{n_{\mathrm{CAD}} + 1/2(n_{\mathrm{FA}} + n_{\mathrm{BD}})}. \quad (111)$$

We can once again use (111) to define its corresponding $\mathrm{AUC}_{\mathrm{F}_1}$ detection measure as

$$\mathrm{AUC}_{\mathrm{F}_1} = 2\frac{\mathrm{AUC}_{\mathrm{ADP}}}{\mathrm{AUC}_{\mathrm{ADP}} + 1/2(\mathrm{AUC}_{\mathrm{ADP}} + \mathrm{AUC}_{\mathrm{BDP}})} \quad (112)$$

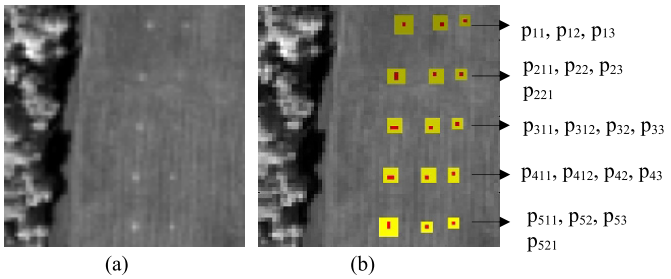


Fig. 13. (a) HYDICE panel scene that contains 15 panels. (b) Ground-truth map of spatial locations of 19 R panel pixels.

which turns out to be also a posteriori 3D ROC curve-derived AUC measure of the F_1-score. As a result, the a priori 3D ROC curve-derived AUC measures presented in this article can be further extended to other applications, such as object detection, including small infrared dim targets and salient/rotated object detection in optical image images not necessarily limited to hyperspectral images. In particular, it is our belief that (110) and (112) can be very effective evaluation metrics for future development in computer vision in objection detection. In other words, a posteriori 3D ROC curve-derived AUC measures can be also derived for small infrared dim TD, such as the BS factor [49], the signal-to-clutter ratio, and the signal-to-clutter ratio gain [49]. Such a posteriori 3D ROC curve-derived AUC measures provide new insight into designing new and better effective detection metrics for object/infrared dim small TD.

Recently, some works have used a box plot [50], [51], referred to as the separability metric map to evaluate the performance of AD and BS performance. For example, Fig. 9(b) in [51] used the gap between box values of detected anomalies and BKG to determine the effectiveness of a detector. In terms of the proposed 3D ROC curve-derived detection measures, the box value of anomalies and the box value of BKG are actually $\mathrm{AUC}_{\mathrm{ADP}}$ and $\mathrm{AUC}_{\mathrm{BDP}}$. The major issue in using the separability metric map is that the anomaly-BKG gap is determined qualitatively by visual inspection but not quantitatively as $\mathrm{AUC}_{\mathrm{ADP}}$ and $\mathrm{AUC}_{\mathrm{BDP}}$. Most importantly, these box plots cannot evaluate joint AD and BS nor AD in BS. For example, in [51], the box value is calculated by excluding the 10% largest and10% smallest values where 10% was an empirical value. The proposed 3D ROC-derived detection measures do not have this issue. They provide quantitative detection performance from all-around aspects.

To facilitate the use of 2D/3D ROC analysis, MATLAB-based software codes developed for calculating AUC and plotting ROC are made available at the website of the Remote Sensing Signal and Image Processing Laboratory (RSSIPL), University of Maryland, Baltimore County (UMBC), https://wiki.umbc.edu/display/rssipl/10.+Download.

VII. Image Data Used for Experiments

A. HYDICE Data Scene

Fig. 13 shows that the image scene to be studied was captured in August 1995 by the aerial hyperspectral digital imagery collection experiment (HYDICE) sensor from a flight

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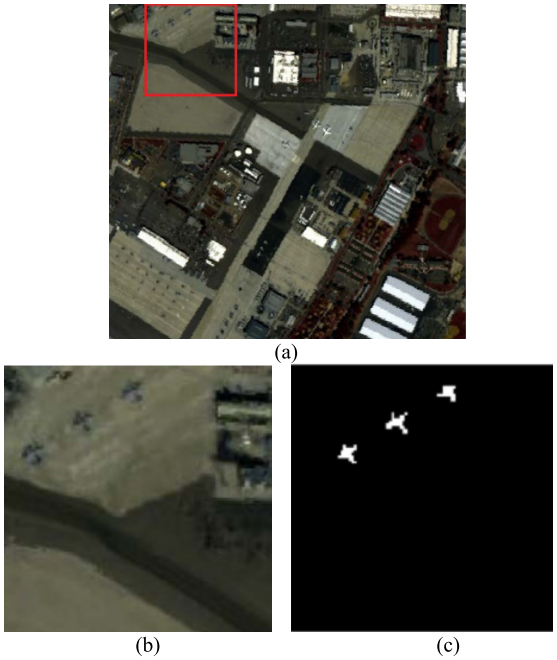


Fig. 14. AVIRIS San Diego Airport scene. (a) Pseudocolor image of the whole scene. (b) Pseudocolor image of the selected area. (c) Ground-truth map.

altitude of 10 000 ft where low signal/high noise bands: bands 1–3 and bands 202–210 and water vapor absorption bands: bands 101–112 and bands 137–153 were removed. The spatial resolution is 1.56 m, and the spectral resolution is 10 nm. This scene has been investigated in depth by many reports, such as [1], [2], [3], and [4].

This scene contains 15 square panels with three different sizes, 3 m × 3 m, 2 m × 2 m, and 1 m × 1 m, respectively, as shown in Fig. 13(a). The ground sampling distance is approximately 1.56 m; the panels in the first column except the first row have two-panel pixels, which are highlighted by red, p_{211} and p_{221} in row 2, p_{311} and p_{312} in row 3, p_{411} and p_{412} in row 4, and p_{511} and p_{521} in row 5, as shown in Fig. 13(b), in the ground truth. All the remaining 11 panels in Fig. 13(b) only contain one single panel pixel for each panel also highlighted by red, p_{11}, p_{12}, and p_{13} in row 1, p_{22} and p_{23} in row 2, p_{32} and p_{33} in row 3, p_{42} and p_{43} in row 4, and p_{52} and p_{53} in row 5. Adding them up, the scene has a total of 19-panel pixels. Their precise spatial locations are shown in Fig. 13(b) with the pixels in green (Y pixels) indicating panel pixels mixed with the BKG.

B. AVIRIS San Diego Airport Scene

Fig. 14(a) shows that a second data scene to be studied is an AVIRIS airport scene from San Diego, CA, USA. The size of the airport is 400 × 400 pixels, and its spatial resolution is 3.5 m with 224 spectral bands, which is the third dimensionality ranging from 370 to 2510 nm. Some bands corresponding to the water absorption regions, the low SNR, and collapse bands (1–6, 33–35, 97, 107–113, 153–166, and 221–224) have been removed and, finally, results in 189 bands. A red square was marked at the upper left corner of the scene, which was chosen as the test image with its ground-truth image shown in Figs. 13(c) and 14(b), respectively. The restricted area has a size of 100 × 100 pixels. The BKG of the scene is mainly made up of a roof, shadow, and grass. The targets that are considered anomalies are three airplanes with the same materials in the scene, which comprised 85 pixels and account for 0.33% of the test image, as shown in Fig. 14(b).

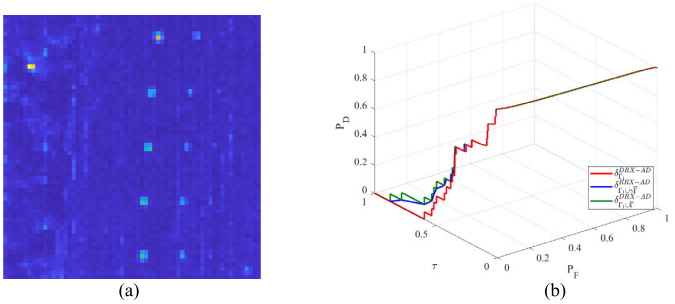


Fig. 15. Detection map of 3D ROC curves produced by RX-AD using three versions, $\delta_{\Gamma_1}^{\text{DRX-AD}}$, $\delta_{\Gamma_1\cup\bar{\Gamma}}^{\text{DRX-AD}}$, and $\delta_{\Gamma_1\cup\gamma\bar{\Gamma}}^{\text{RRX-AD}}$, for the HYDICE panel scene. (a) Detection map. (b) 3D ROC curves.

VIII. Comprehensive Analysis of ROC Curves

In order to demonstrate differences between the traditional ROC curve of (P_D,P_F) and the ROC curve calculated by Theorems 1 and 2, we conducted experiments to address two scenarios. For simplicity of illustration, only the most commonly used anomaly detector developed by Reed and Yu [3], denoted by RX-AD, was implemented for experiments. Nevertheless, all the following conducted analyses are readily applied to any other anomaly detector.

A. True ROC Curves Produced by DNPD and RNPD

The first one is to calculate true ROC curves where two DNPDs, $\delta_{\Gamma_1}^{\text{DNPD}}$ and $\delta_{\Gamma_1\cup\bar{\Gamma}}^{\text{DNPD}}$, and RNPD, $\delta_{\Gamma_1\cup\gamma\bar{\Gamma}}^{\text{RNPD}}$, plotted for a hyperspectral image, which uses given data samples, $\Gamma = \{\mathbf{r}_i\}_{i=1}^{N}$, as threshold values. This is the true scenario without using unavailable data samples to linearly interpolate ROC curves by an NP randomized detector. The calculated $\text{AUC}_{(\text{D,F})}$ are $\text{AUC}_{\Gamma_1}^{\text{DNPD}}(\text{D,F})$, $\text{AUC}_{\Gamma_1\cup\bar{\Gamma}}^{\text{DNPD}}(\text{D,F})$, and $\text{AUC}_{\Gamma_1\cup\gamma\bar{\Gamma}}^{\text{RNPD}}(\text{D,F})$ by Theorem 2 according to Fig. 4 where $\text{AUC}_{\Gamma_1\cup\gamma\bar{\Gamma}}^{\text{RNPD}}(\text{D,F})$ is the linearly interpolated piecewise ROC produced by RNPD between $\text{AUC}_{\Gamma_1}^{\text{DNPD}}(\text{D,F})$ and $\text{AUC}_{\Gamma_1\cup\bar{\Gamma}}^{\text{DNPD}}(\text{D,F})$.

When $\delta_{\Gamma_1}^{\text{DNPD}}$ in (37) is implemented, its ROC curve starts at $(P_D,P_F) = (0,0)$ but does not end at $(P_D,P_F) = (1,1)$. On the other hand, as $\delta_{\Gamma_1\cup\bar{\Gamma}}^{\text{DNPD}}$ is in (38) implemented, its ROC curve does not start at $(P_D,P_F) = (0,0)$ but does end at $(P_D,P_F) = (1,1)$. For both cases, their AUC values need to be normalized, as discussed in [7]. It is only $\delta_{\Gamma_1\cup\gamma\bar{\Gamma}}^{\text{RNPD}}$ in (39), which can start at $(P_D,P_F) = (0,0)$) but also end at $(P_D,P_F) = (1,1)$ due to its use of linear interpolation by RNPD. Thus, to faithfully compare $\text{AUC}_{\Gamma_1\cup\gamma\bar{\Gamma}}^{\text{DNPD}}(\text{D,F})$, $\text{AUC}_{\Gamma_1}^{\text{DNPD}}(\text{D,F})$ and $\text{AUC}_{\Gamma_1\cup\bar{\Gamma}}^{\text{DNPD}}(\text{D,F})$ are not normalized to reflect their true values in the following experiments.

Fig. 15(a) shows the detection map produced by RX-AD. Fig. 15(b) plots the red, green, and blue 3D ROC curves

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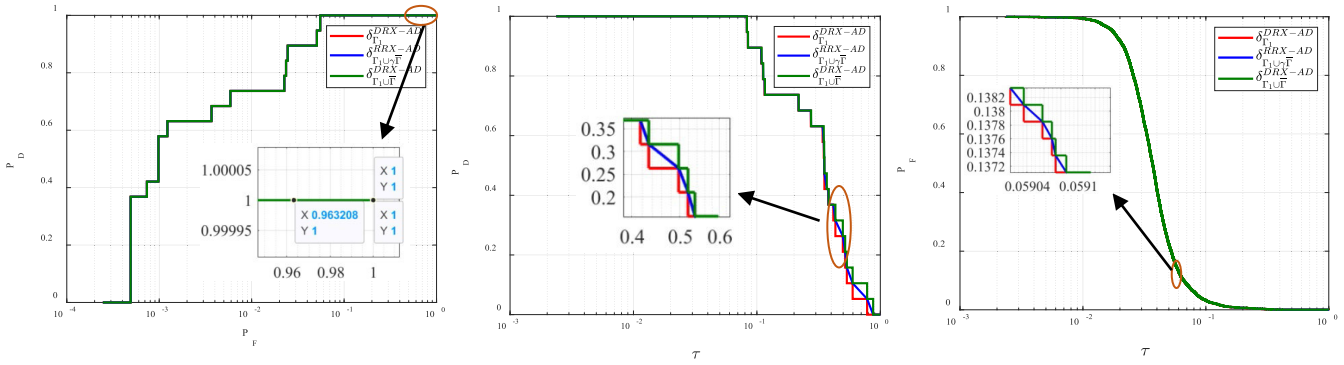


Fig. 16. Three 2D ROC curves of (P_D,P_F), (P_D,τ), and (P_F,τ) generated from the 3D ROC curves in Fig. 15. (a) 2D ROC curves of (P_D,P_F). (b) 2D ROC curves of (P_D,τ). (c) 2D ROC curves of (P_F,τ).

TABLE II
AUC VALUES PRODUCED BY RX-AD USING THREE VERSIONS, $\delta_{\Gamma_1}^{\text{DRX-AD}}$, $\delta_{\Gamma_1\cup\bar{\Gamma}}^{\text{DRX-AD}}$, AND $\delta_{\Gamma_1\cup\gamma\bar{\Gamma}}^{\text{RRX-AD}}$, FOR THE HYDICE PANEL SCENE IN FIG. 16

detector	$AUC_{(D,F)}$	AUC_{ADP}	AUC_{BDP}	AUC_{JAD}	AUC_{JBS}	AUC_{ADBS}	AUC_{SNPR}	AUC_{OADP}
$\delta_{\Gamma_1}^{\text{DRX-AD}}$	0.989518	0.344455	0.957106	1.333973	1.946624	1.301561	8.030405	2.291079
$\delta_{\Gamma_1\cup\bar{\Gamma}}^{\text{DRX-AD}}$	0.989763	0.370376	0.956982	1.360138	1.946745	1.327357	8.609714	2.31712
$\delta_{\Gamma_1\cup\gamma\bar{\Gamma}}^{\text{RRX-AD}}$	0.989763	0.357415	0.957044	1.347178	1.946807	1.314459	8.320479	2.304222

produced by RX-AD using three versions, $\delta_{\Gamma_1}^{\text{DRX-AD}}$, $\delta_{\Gamma_1\cup\bar{\Gamma}}^{\text{DRX-AD}}$, and $\delta_{\Gamma_1\cup\gamma\bar{\Gamma}}^{\text{RRX-AD}}$ for the HYDICE panel scene, respectively, where it confirms Theorems 2 and 3 to show that the $\delta_{\Gamma_1\cup\gamma\bar{\Gamma}}^{\text{RRX-AD}}$-generated blue 3D ROC curve is, indeed, sandwiched between the $\delta_{\Gamma_1}^{\text{DRX-AD}}$-generated red 3D ROC curve and the $\delta_{\Gamma_1\cup\bar{\Gamma}}^{\text{DRX-AD}}$-generated green 3D ROC curve.

As expected, the red 3D ROC curve starts at (0,0) but does not end at (1,1) as opposed to the green 3D ROC curve, which does not start at (0,0) but does end at (1,1). Nevertheless, the $\delta_{\Gamma_1\cup\gamma\bar{\Gamma}}^{\text{RRX-AD}}$-generated blue 3D ROC curve is the only one starting at (0,0) and ending at (1,1).

Fig. 16(a)–(c) further shows the three 2D ROC curves of (P_D,P_F), (P_D,τ), and (P_F,τ) generated from the 3D ROC curves in Fig. 15 where a particular small range of each 2D ROC curve is selected and circled to magnify their values. As shown in Fig. 16, it is interesting to note that the three 2D ROC curves of (P_D,P_F) are nearly identical, while the 2D ROC curves of (P_D,τ) shows the largest visible discrepancy among all the three NPD: $\delta_{\Gamma_1}^{\text{DRX-AD}}$, $\delta_{\Gamma_1\cup\bar{\Gamma}}^{\text{DRX-AD}}$, and $\delta_{\Gamma_1\cup\gamma\bar{\Gamma}}^{\text{RRX-AD}}$. The 2D ROC curves of ($P_F$,$\tau$) is right in between. This intriguing finding actually confirms a long-standing issue of why plotting 2D ROC curve of (P_D,P_F) has never been a concern. This is because P_D and P_F are plotted by the same threshold τ, and thus, the 2D ROC curves generated by three detectors, $\delta_{\Gamma_1}^{\text{DRX-AD}}$, $\delta_{\Gamma_1\cup\bar{\Gamma}}^{\text{DRX-AD}}$, and $\delta_{\Gamma_1\cup\gamma\bar{\Gamma}}^{\text{RRX-AD}}$, are nearly the same with no appreciable differences. Unfortunately, this is not true for the other two 2D ROC curves that are, in fact, plotted based on the independent parameter τ. These simple experiments demonstrated several fundamental issues. One is that the 2D ROC curve of (P_D,P_F) cannot be used to effectively evaluate target detectability and BKG suppressibility. Another is that solely using 2D ROC curve of (P_D,P_F) as an evaluation criterion is misleading. Specifically, most 2D ROC curves of (P_D,P_F) reported in the literature are actually plotted as step functions by DNPD either $\delta_{\Gamma_1}^{\text{DRX-AD}}$ or $\delta_{\Gamma_1\cup\bar{\Gamma}}^{\text{DRX-AD}}$ but not RNPD $\delta_{\Gamma_1\cup\gamma\bar{\Gamma}}^{\text{RRX-AD}}$. Since 2D ROC curves of (P_D,P_F) generated by these three NPDs are almost the same, it never flags an issue that these three detectors are in fact distinct.

To further conduct a quantitative study and analysis, Table II tabulates the AUC values based on $AUC_{(D,F)}$ along with seven detection measures, AUC_{ADP}, AUC_{BDP}, AUC_{JAD}, AUC_{JBS}, AUC_{ADBS}, AUC_{SNPR}, and AUC_{OADP}, for quantitative comparison where $AUC_{(D,F)}$ values calculated by the three NPDs were so close within a tolerance of 10^{-3}, and they are indistinguishable. By contrast, the $AUC_{(D,\tau)}$ values had the largest differences among the 2D ROC curves of (P_D,P_F), (P_D,τ), and (P_F,τ). Despite that some recent works begin to use the AUC values of 2D ROC curves of (P_D,P_F), (P_D,τ), and (P_F,τ) to evaluate AD performance, it seems that these three criteria may not be sufficient. This is because an effective measure to evaluate an anomaly detector must take both AD and BS into account not only based on $AUC_{(D,F)}$, AUC_{ADP}, and AUC_{BDP}. By looking into all these eight detection measures, the greatest difference came from SNPR followed by JAD and ADBS. All these three measures are specifically designed to jointly assess anomaly detectability and BKG suppressibility.

Similarly, the same experiments performed for the HYDICE panel scene were also conducted for the San Diego airport scene. Fig. 17(a) shows the detection map produced by

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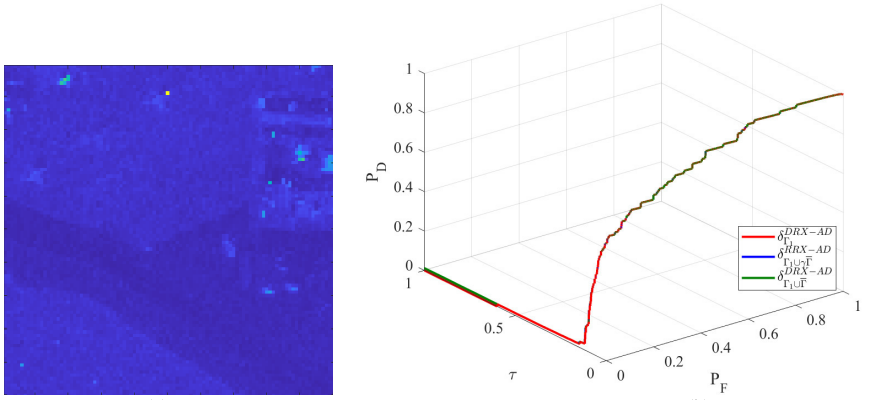


Fig. 17. (a) Detection map of (b) 3D ROC curves produced by RX-AD using three versions, $\delta_{\Gamma_1}^{\text{DRX-AD}}$, $\delta_{\Gamma_1\cup\bar{\Gamma}}^{\text{DRX-AD}}$, and $\delta_{\Gamma_1\cup\gamma\,\bar{\Gamma}}^{\text{RRX-AD}}$ for the San Diego airport scene.

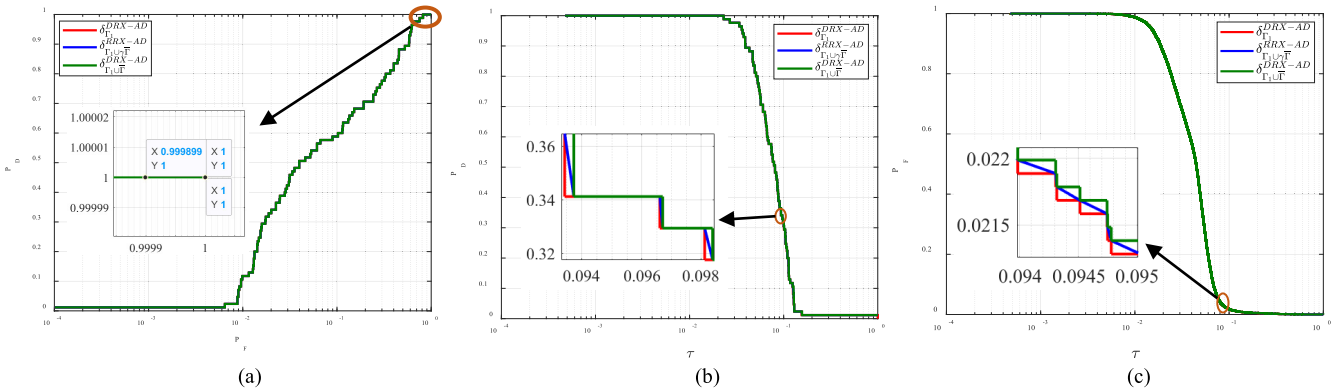


Fig. 18. Three 2D ROC curves of (P_D,P_F), (P_D,τ), and (P_F,τ) generated from the 3D ROC curves in Fig. 17. (a) 2D ROC curves of (P_D,P_F). (b) 2D ROC curves of (P_D,τ). (c) 2D ROC curves of (P_F,τ).

TABLE III

AUC VALUES PRODUCED BY RX-AD USING THREE VERSIONS, $\delta_{\Gamma_1}^{\text{DRX-AD}}$, $\delta_{\Gamma_1\cup\bar{\Gamma}}^{\text{DRX-AD}}$, AND $\delta_{\Gamma_1\cup\gamma\,\bar{\Gamma}}^{\text{RRX-AD}}$, FOR THE SAN DIEGO AIRPORT SCENE IN FIG. 18

detector	$\text{AUC}_{(D,F)}$	AUC_{ADP}	AUC_{BDP}	AUC_{JAD}	AUC_{JBS}	AUC_{ADBS}	AUC_{SNPR}	AUC_{OADP}
$\delta_{\Gamma_1}^{\text{DRX-AD}}$	0.83327	0.09092	0.95477	0.92419	1.78805	1.04570	2.0106	1.87897
$\delta_{\Gamma_1\cup\bar{\Gamma}}^{\text{DRX-AD}}$	0.83337	0.09108	0.95467	0.92446	1.78805	1.04575	2.00950	1.879136
$\delta_{\Gamma_1\cup\gamma\bar{\Gamma}}^{\text{RRX-AD}}$	0.83338	0.09100	0.95472	0.92438	1.78810	1.04573	2.01007	1.87911

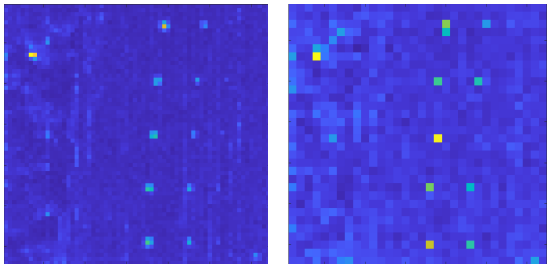
RX-AD. Fig. 17(b) plots the red, green, and blue 3D ROC curves produced by RX-AD using three versions: $\delta_{\Gamma_1}^{\text{DRX-AD}}$, $\delta_{\Gamma_1\cup\bar{\Gamma}}^{\text{DRX-AD}}$ and $\delta_{\Gamma_1\cup\gamma\,\bar{\Gamma}}^{\text{RRX-AD}}$. Unlike Fig. 15, the three 3D ROC curves are so close together, and it is extremely difficult to tell them apart.

Fig. 18(a)–(c) also shows similarly that the three 2D ROC curves of (P_D,P_F), (P_D,τ), and (P_F,τ) generated from the 3D ROC curves in Fig. 17 are very close with no visible difference where a particular small range of each 2D ROC curve is selected and circled to magnify the values. This indicates that there are no appreciable differences among using the three NPDs. Table III tabulates the AUC values according to AUC_{ADP}, AUC_{BDP}, AUC_{JAD}, AUC_{JBS}, AUC_{ADBS}, AUC_{SNPR}, and AUC_{OADP} for quantitative comparison. In particular, except AUC_{ADP}, all the other seven AUC values were so close within a tolerance of 10^{-3} that there was no difference among using the three detectors.

B. ROC Curves Produced by Insufficient Sampling

The second one is the case that data samples are available but are not all used to plot an ROC curve. In other words, not every data sample is used to calculate (P_D,P_F). In this case, an insufficient sampling (IS) is used to extract a partial subset of the hyperspectral image from $\Gamma = \{\mathbf{r}_i\}_{i=1}^{N}$ and $\Gamma^{\text{IS}} = \{\mathbf{r}_i\}_{i=1}^{n_{\text{IS}}}$ to plot ROC curves. This scenario is designed to see how close such insufficient sampled ROC curve by linear interpolation to its corresponding $\text{AUC}_{\Gamma_1}^{\text{DNPD}}$(D,F) and $\text{AUC}_{\Gamma_1\cup\bar{\Gamma}}^{\text{DNPD}}$(D,F). Three scenarios are of interest.

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(a) (b)

Fig. 19. Detection maps of the HYDICE panel scene by RX-AD using two different sizes. (a) Original size (64 × 64). (b) DS size by 2 (32 × 32).

1) Downsampled Dataset: One scenario is to use down sampling to obtain a partial set of data samples to perform HAD and then evaluate its ROC curve performance against that using full dataset for comparative analysis. Examples include multiresolution analysis and superpixel segmentation with downsampled (DS) data samples. For the purpose of illustration, we only used the HYDICE panel scene as an example for experiments.

There are 4096 pixels in the image of size 64 × 64 in Fig. 13, which are numbered from left to right and top to bottom. Now, we implemented DS by sampling every 4 pixels to yield an image of size 32 × 32, which results in $n_{\mathrm{IS}} = 1024$ pixels. Fig. 19(a) and (b) show two different sizes of the HYDICE panel scene with the original data size in Fig. 19(a) and the reduced size shown in Fig. 19(b) which was obtained using DS by 2. It should be noted that Fig. 19(b) only used one-quarter of the data samples in Fig. 19(a) to form Fig. 19(b).

Comparing Fig. 19(a) and Fig. 19(b), it clearly showed that downsampling increased BKG uncertainty and also showed its effect on smearing and blurring panel pixels, particularly those in the second and third columns. Fig. 20 plots 3D ROC curves produced by RX-AD with three versions, $\delta_{\Gamma_1}^{\text{DRX-AD}}$, $\delta_{\Gamma_1\cup\bar{\Gamma}}^{\text{DRX-AD}}$, and $\delta_{\Gamma_1\cup\gamma\,\bar{\Gamma}}^{\text{RRX-AD}}$, using the DS size of 1024 pixels in Fig. 19(b).

Comparing Fig. 15 and Fig. 20, the differences among the three 3D ROC curves are more visible where the green 3D ROC curve generated by $\delta_{\Gamma_1\cup\bar{\Gamma}}^{\text{DRX-AD}}$ clearly did not start (0,0), while the other two did.

Fig. 21(a)–(c) shows three 2D ROC curves of (P_D,P_F), (P_D,τ), and (P_F,τ) calculated from the 3D ROC curves in Fig. 20 where a particular small range of each 2D ROC curve is selected and circled to magnify their values. In analogy with Fig. 16, the three 2D ROC curves of (P_D,P_F) in Fig. 21(a) were very close as expected. However, the three 2D ROC curves of (P_D,τ) in Fig. 21(b) have shown great discrepancy as τ became large. However, this phenomenon was not visible in the 2D ROC curves of (P_F,τ) in Fig. 21(c).

Table IV tabulates the AUC values of the ROC curves in Fig. 21 according to AUC_{ADP}, AUC_{BDP}, AUC_{JAD}, AUC_{JBS}, AUC_{ADBS}, AUC_{SNPR}, and AUC_{OADP} for quantitative comparison where some interesting observations are worth being noted in comparison with Table I.

First, $\text{AUC}_{(\text{D,F})}$ values did not matter very much caused by IS with their differences within 10^{-3}. By contrast, the AUC_{ADP} values caused by IS were significantly increased from the

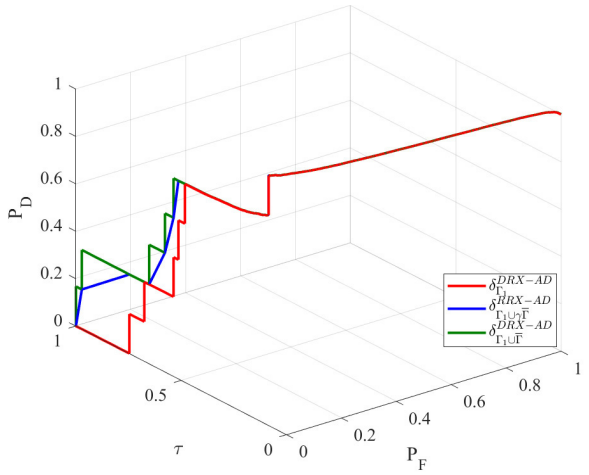


Fig. 20. 3D ROC curves produced by RX-AD with three versions, $\delta_{\Gamma_1}^{\text{DRX-AD}}$, $\delta_{\Gamma_1\cup\bar{\Gamma}}^{\text{DRX-AD}}$, and $\delta_{\Gamma_1\cup\gamma\,\bar{\Gamma}}^{\text{RRX-AD}}$, using DS size of 1024 pixels in Fig. 19(b).

range of 0.3–0.37 to the range of 0.58–0.65, a significant jump of 0.28. Interestingly, BDP performed completely opposite with AUC_{BDP} values significantly decreased from the range of 0.956–0.957 down to the range of 0.909–0.910 with a drop of 0.05. Thus, $\text{AUC}_{(\text{D,F})}$ could not suffice to be used to evaluate the differences, while AUC_{ADP} and AUC_{BDP} could. If we further look into the other five detection measures, they did better than $\text{AUC}_{(\text{D,F})}$ and AUC_{BDP} in the assessment of differences between the original image and insufficient sampled image data. One most interesting finding is that not all eight detection measures agree on detection performance assessment. Since $\text{AUC}_{(\text{D,F})}$ is not useful, we group the other seven detection measures into two categories. One category is made up of AUC_{BDP}, AUC_{JBS}, and AUC_{SNPR} that are closely related to the BKG assessment where the performance using the original data did better than its insufficient sampled data. Conversely, the other category is composed of AUC_{ADP}, AUC_{JAD}, AUC_{ADBS}, and AUC_{OADP} that are closely related to anomaly detectability where performance using the original data did worse than its insufficient sampled data. From these experiments, we can see that evaluating anomaly detectability and BKG suppressibility are separate issues and cannot be evaluated by a single detection measure, such as $\text{AUC}_{(\text{D,F})}$, which has been shown to be ineffective.

2) Partial Data Subset With Unavailable Data Samples: The second scenario is to also use a partial set of data samples to calculate ROC curves from the detection map produced by the original entire data samples. This scenario is different from the first scenario presented in Section VIII-B1 in the sense that the ROC curves are calculated directly from the original detection map produced by RX-AD on the original entire dataset. This issue is particularly important and crucial. It generally arises from medical data analysis due to practical constraints with missing data samples.

To simulate this study, we consider the HYDICE dataset in Fig. 13 as the full dataset with 4096 data samples, denoted by $\mathbf{X}$. However, only 1024 data samples are available to be used to plot ROC curves. These 1024 data samples were

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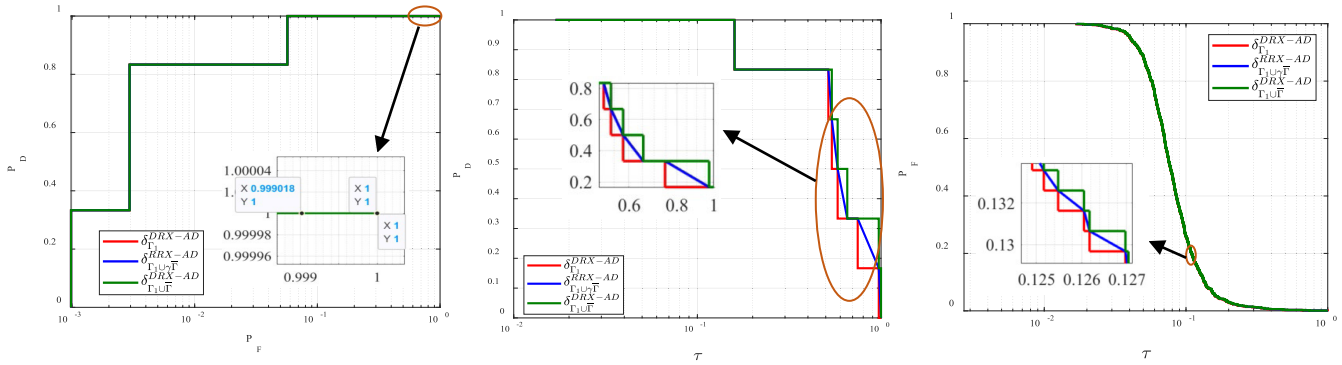


Fig. 21. 3D ROC curves and their corresponding three 2D ROC curves of (P_D,P_F), (D,τ), and (F,τ) produced by RX-AD with three versions, $\delta_{\Gamma_1}^{\text{DRX-AD}}$, $\delta_{\Gamma_1\cup\bar{\Gamma}}^{\text{DRX-AD}}$, and $\delta_{\Gamma_1\cup\gamma\bar{\Gamma}}^{\text{RRX-AD}}$, using IS of 1024 pixels for the HYDICE panel scene.

TABLE IV

AUC VALUES PRODUCED BY RX-AD USING THREE VERSIONS, $\delta_{\Gamma_1}^{\text{DRX-AD}}$, $\delta_{\Gamma_1\cup\bar{\Gamma}}^{\text{DRX-AD}}$, AND $\delta_{\Gamma_1\cup\gamma\bar{\Gamma}}^{\text{RRX-AD}}$, FOR THE HYDICE PANEL SCENE IN FIG. 20

detector	$\text{AUC}_{(\text{D,F})}$	AUC_{ADP}	AUC_{BDP}	AUC_{JAD}	AUC_{JBS}	AUC_{ADBS}	AUC_{SNPR}	AUC_{OADP}
$\delta_{\Gamma_1}^{\text{DRX-AD}}$	0.987721022	0.58578803	0.910225004	1.573509051	1.897946025	1.496013033	6.525068821	2.483734055
$\delta_{\Gamma_1\cup\bar{\Gamma}}^{\text{DRX-AD}}$	0.98870334	0.651114646	0.909627715	1.639817986	1.898331054	1.560742361	7.20480447	2.549445701
$\delta_{\Gamma_1\cup\gamma\bar{\Gamma}}^{\text{RRX-AD}}$	0.98870334	0.618451338	0.909926359	1.607154678	1.898629699	1.528377697	6.866063498	2.517081037

TABLE V

AUC VALUES PRODUCED BY RX-AD WITH THREE VERSIONS, $\delta_{\Gamma_1}^{\text{DRX-AD}}$, $\delta_{\Gamma_1\cup\bar{\Gamma}}^{\text{DRX-AD}}$, AND $\delta_{\Gamma_1\cup\gamma\bar{\Gamma}}^{\text{RRX-AD}}$, USING 4096 AND 1024 PIXELS FOR THE HYDICE PANEL SCENE IN FIG. 22

detector	$\text{AUC}_{(\text{D,F})}$	AUC_{ADP}	AUC_{BDP}	AUC_{JAD}	AUC_{JBS}	AUC_{ADBS}	AUC_{SNPR}	AUC_{OADP}
$\delta_{\Gamma_1}^{\text{DRX-AD}}$ with $\mathbf{X}$	**0.989518**	0.344455	0.957106	1.333973	1.946624	1.301561	8.030405	2.291079
$\delta_{\Gamma_1}^{\text{DRX-AD}}$ with $\mathbf{X}_{\text{subset}}$	0.989194	**0.384713**	**0.958717**	**1.373908**	**1.947912**	**1.343431**	**9.319028**	**2.332625**
$\delta_{\Gamma_1\cup\bar{\Gamma}}^{\text{DRX-AD}}$ with $\mathbf{X}$	0.989763	0.370376	0.956982	1.360138	1.946745	1.327357	8.609714	2.31712
$\delta_{\Gamma_1\cup\bar{\Gamma}}^{\text{DRX-AD}}$ with $\mathbf{X}_{\text{subset}}$	**0.990177**	**0.488851**	**0.958349**	**1.479028**	**1.948526**	**1.4472**	**11.73681**	**2.437377**
$\delta_{\Gamma_1\cup\gamma\bar{\Gamma}}^{\text{RRX-AD}}$ with $\mathbf{X}$	0.989763	0.357415	0.957044	1.347178	1.946807	1.314459	8.320479	2.304222
$\delta_{\Gamma_1\cup\gamma\bar{\Gamma}}^{\text{RRX-AD}}$ with $\mathbf{X}_{\text{subset}}$	**0.990177**	**0.436782**	**0.958533**	**1.426959**	**1.94871**	**1.395315**	**10.53329**	**2.385492**

obtained by sampling every 4 pixels to yield a partial data subset, denoted by $\mathbf{X}_{\text{subset}}$, in which case there were 3 pixels missing when each pixel was sampled in the DS dataset. Unlike the first scenario in Section VIII-B1 where the DS dataset was used to perform AD, this section considers the partial data subset, $\mathbf{X}_{\text{subset}}$, as the original dataset with missing data samples. Thus, the detection map produced by the full dataset was used for both $\mathbf{X}$ and $\mathbf{X}_{\text{subset}}$ to calculate ROC curves for comparison. In this case, the discrepancy of ROC curves between using $\mathbf{X}$ and $\mathbf{X}_{\text{subset}}$ with missing data samples was quantitatively analyzed.

Fig. 22(a)–(d) plots 3D ROC curves and their corresponding 2D ROC curves generated from the detection map in Fig. 19(a) by $\mathbf{X}$ and $\mathbf{X}_{\text{subset}}$ where solid lines and dashed lines are curves generated by $\mathbf{X}$ and $\mathbf{X}_{\text{subset}}$, respectively. Table V tabulates the AUC values according to AUC_{ADP}, AUC_{BDP}, AUC_{JAD}, AUC_{JBS}, AUC_{ADBS}, AUC_{SNPR}, and AUC_{OADP} for quantitative comparison where the best results are boldfaced in comparison between two datasets. Except for only one case of $\text{AUC}_{(\text{D,F})}$ by $\delta_{\Gamma_1}^{\text{DRX-AD}}$ using $\mathbf{X}$, all other results obtained by $\mathbf{X}_{\text{subset}}$ were better than that obtained by using $\mathbf{X}$ across aboard.

The above experiments demonstrated that, using a partial set of data samples, $\mathbf{X}_{\text{subset}}$ could do better than using the full dataset $\mathbf{X}$. This indicates that conclusions drawn from $\mathbf{X}_{\text{subset}}$ may sometimes be misleading.

3) Uniform Sampled Data: The third scenario arises from uniform sampling to plot ROC curves. It uses the uniform step size, as discussed in [7], to sample the values of P_F to plot an ROC curve, denoted by $\text{AUC}_{(\text{D,F})}^{\text{uniform}}$. This is a special case of a partial data subset in Section VIII-B2 but has an additional issue due to the fact that, for each β, for example,

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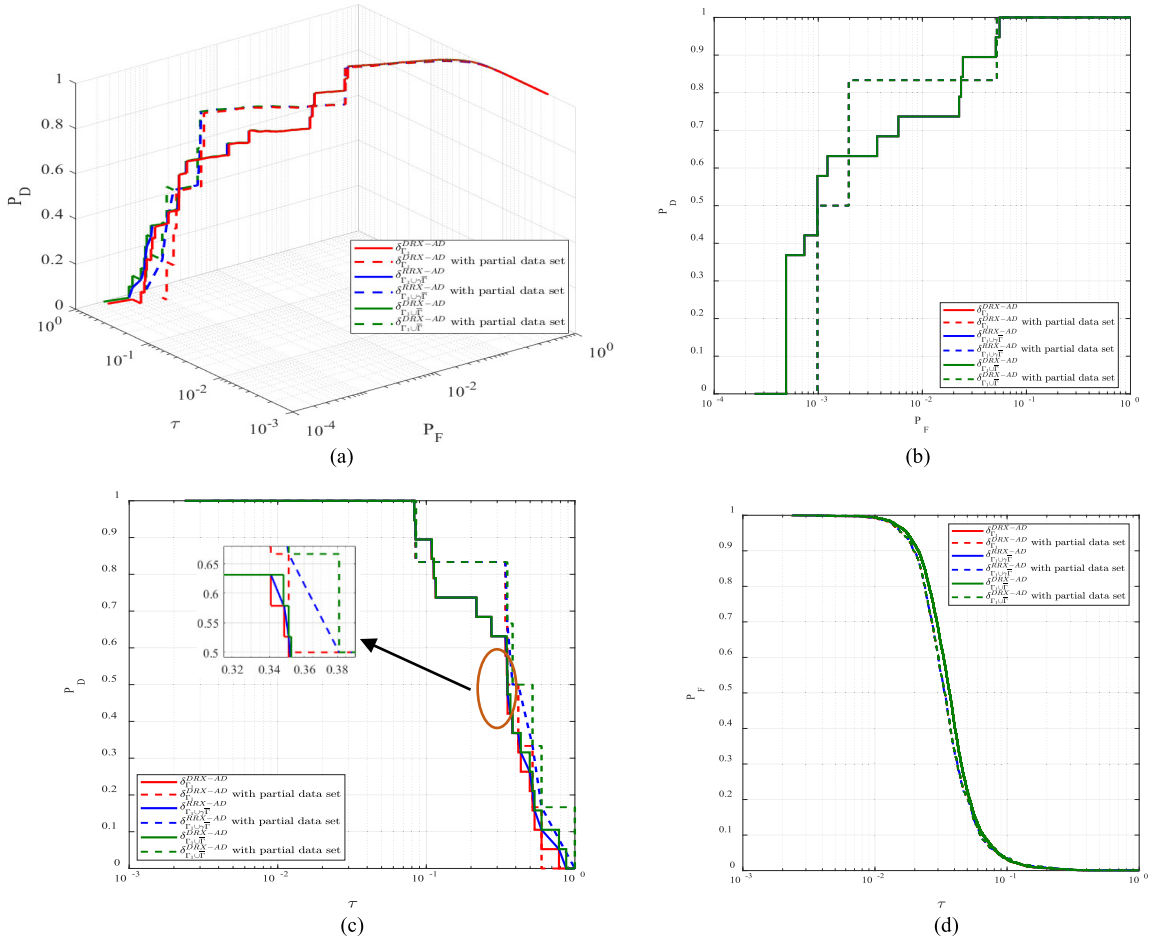


Fig. 22. 3D ROC curves and their corresponding 2D ROC curves generated from the detection map in Fig. 19(a) by $\mathbf{X}$ and$\mathbf{X}_{\mathrm{DS}}$. (a) 3D ROC curves. (b) 2D ROC curves of ($\mathrm{P_D}$,$\mathrm{P_F}$). (c) 2D ROC curves of ($\mathrm{P_D}$,τ). (d) 2D ROC curves of ($\mathrm{P_F}$,τ).

$\beta = 10^{-1}$, it is very likely that no data samples can produce $\mathrm{P_F}$ corresponding to $\beta = 10^{-1}$. This is often the case when a comparative performance analysis is evaluated based on the value of $\mathrm{P_D}$ with a fixed value of β or on the value of $\mathrm{P_F}$ by fixing $\mathrm{P_D}$ at a certain value, such as experiments in [20]. To mitigate this dilemma, we need to appeal for the implicit threshold parameter τ via (7) and use it to produce its pair of ($\mathrm{P_D}$,$\mathrm{P_F}$). However, under this circumstance, there are more than two data samples that produce pairs of ($\mathrm{P_D}$,$\mathrm{P_F}$) values between two consecutive threshold values of τ by uniform sampling. When this occurs, using the linear interpolation is not accurate since there are true data samples that can actually produce real values of ($\mathrm{P_D}$,$\mathrm{P_F}$) since this scenario has been discussed with extensive experiments and analysis in [7], and will not be included here but rather referred to [7] for all details.

The experiments conducted in the section demonstrate that various detection measures yield different conclusions. This indicates that when it comes to AD, the anomaly detectability, BKG suppressibility, and joint detectability of anomalies in BKG must be all factored into the evaluation to provide an all-around analysis. Apparently, the eight detection measures developed in this article offer such a need.

IX. Conclusion

It is well-known that RX-AD in [3] is derived from GLRT in [4]. This result has been taken for granted for years. However, it requires substantial knowledge of binary hypothesis testing theory to fully understand how such GLRT → RX-AD derivation works. Most recently, Chang [52] derived a theory for how to convert targets to anomalies for AD. Since some basic issues were not investigated in the literature, Chang [8] rederived RX-AD without using GLRT. Interestingly, similar issues that occurred in the GLRT → RX-AD derivation are also found in applying the 2D ROC curve of ($\mathrm{P_D}$,$\mathrm{P_F}$) to hyperspectral detection, which has been taken for granted as well. Several fundamental and conceptual issues have been overlooked and ignored but have very important implications in concept understanding, particularly, in calculating the AUC value of a 2D ROC curve of ($\mathrm{P_D}$,$\mathrm{P_F}$) and using it to evaluate AD performance. This article addresses these issues and conducts a comprehensive study on the mathematical analysis of the ROC curve from the NP detection theory with detailed mathematical proofs. Specifically, two issues associated with an anomaly detector are investigated. One is that a hyperspectral anomaly detector is, in fact, an RNPD implemented on a discrete set of data samples. The other is that the 2D ROC curve particularly designed to evaluate TD

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may not work effectively for AD since anomaly detector is an unknown TD, and BKG suppression is a crucial issue.

When an anomaly detector works as an RNPD, it makes a deterministic detection on each of the given data samples, but random detections on those data samples are not provided by the dataset. This article offers a rigorous mathematical treatment of how a 2D ROC curve is derived by RNPD and further proves that the resulting 2D ROC curve is actually a step function, which can be linearly piecewise interpolated by RNPD for P_D and P_F values for unavailable data samples between discrete hyperspectral data samples. Over the past years, the $AUC_{(D,F)}$ value of a 2D ROC curve of (P_D,P_F) has been widely used for AD performance evaluation. This article provides hard evidence that conclusions solely based on $AUC_{(D,F)}$ to evaluate an anomaly detector are, indeed, misleading. Thus, the main contributions are the details of mathematical proofs and derivations that are key to understanding how an ROC curve is generated.

Unlike TD that detects known targets in noise, AD considers BKG as a crucial factor in performance evaluation. Thus, to better address this issue, AD problem is reformulated a detection problem of a two-class classification problem specified by anomaly and BKG via a confusion matrix that can be used to calculate anomaly detectability and BKG suppressibility. In particular, we revisit a 3D ROC curve and rederive eight detection measures by a confusion matrix to take care of anomaly detectability, BKG suppressibility, and joint anomaly detectability with BKG suppressibility for AD. It turns out that these measures offer all-around performance evaluation for AD, which can better assess many AD methods published in the literature.

Although many issues were noted in [7], they were neither addressed nor investigated in [7] due to its limited space. This article resolved these issues. Particularly, the issue of ROC curves as step functions along with its linear piecewise interpolation by RNPD on unavailable data samples was not explored and discussed in [7]. Accordingly, this article supplements what was missed in [7] and can be considered to be a companion paper as its sequel.

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